



UNIVERSITY OF SAN CARLOS
SCHOOL OF ENGINEERING

PHYSICS ENFORCEMENT ON MACHINE
LEARNING SURROGATES OF THE ACTIVATED
SLUDGE PROCESS
DISSERTATION

2026

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Study Programme	Doctor of Engineering (D.Eng.)
Field of Study	Water Engineering
Department	Department of Industrial Engineering
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Cebu City, Philippines, 2026
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Declaration

I hereby declare that the work presented in the dissertation entitled “Physics Enforcement on Machine Learning Surrogates of the Activated Sludge Process” is original and the result of my own research. Formulations and ideas taken from other sources are cited as such.

Cebu City, Philippines, 2026

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Acknowledgments

The author gratefully acknowledges the Department of Science and Technology (DOST) of the Philippines for supporting his doctoral studies from 2022 to 2026 through the Engineering Research and Development for Technology (ERDT) scholarship. He is deeply grateful to his adviser, Dr. Lorafe F. Lozano, for her guidance and unwavering support throughout this research. He also thanks Ms. Alele Lagusan of the USC-ERDT office for her invaluable assistance with the administrative matters of his studies as an ERDT scholar at USC. The author is likewise indebted to Engr. Luis Cabatingan, his closest batchmate in the D.Eng. program, whose stimulating intellectual conversations on chemical engineering helped inspire the topics explored in this dissertation. Finally, he thanks his boyfriend, Leomark, for the patience, love, and steadfast support that sustained him physically and emotionally throughout his doctoral journey.

Abstract

Biological wastewater treatment relies on activated sludge processes whose behavior is governed by interacting carbon, nitrogen, phosphorus, oxygen, biomass, and solids transformations. Mechanistic models can represent these interactions, but repeated simulation can be burdensome when many operating and influent conditions must be examined. This research develops a surrogate modeling framework for activated sludge prediction that combines machine learning accuracy, physical admissibility, and interpretability. The study first evaluates whether modern tabular learning models can approximate the complete component response of a mechanistic activated sludge simulator from paired input and output data. It then examines how post prediction enforcement of conservation and nonnegativity affects surrogate outputs, and it considers how interpretable constrained regression can support engineering judgment. The work is scoped to simulated steady-state data under a defined activated sludge component representation. Within that scope, the framework clarifies how fast data driven prediction can be evaluated alongside the physical meaning of the returned component state.

Keywords. activated sludge process; machine learning surrogate; wastewater treatment; physical admissibility; interpretable modeling

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List of Algorithms

List of Abbreviations

ASM	activated sludge model	nAUC	normalized area under the learning curve
ASM2d-TSN	Activated Sludge Model No. 2d with two-step nitrification	nMAE	normalized mean absolute error
COD	chemical oxygen demand	nMSE	normalized mean squared error
HRT	hydraulic retention time	nRMSE	normalized root mean squared error
ICSOR	Invariant-Constrained Second-Order Regression	PLS	partial least squares
k-NN	k-nearest neighbors	RMSE	root mean squared error
MAE	mean absolute error	SVR	support vector regression
MAPE	mean absolute percentage error	TN	total nitrogen
MLP	multilayer perceptron	TP	total phosphorus
MSE	mean squared error	TSS	total suspended solids

List of Symbols

$\mathbb{R}$	real numbers
$\mathbb{N}$	nonnegative integers
$\ \mathbf{u}\ _p$	L^p -norm of $\mathbf{u}$
$\ \cdot\ _2$	Euclidean norm
$\ \cdot\ _1$	L^1 -norm (sum of absolute values)
$\ \cdot\ _F$	Frobenius norm of a matrix
$(\cdot)^\top$	transpose of a vector or matrix
$\otimes$	Kronecker product
$\oslash$	componentwise (Hadamard) division
$\text{Diag}(\cdot)$	diagonal matrix formed from a vector
I_F	identity matrix of order F
$\mathbf{1}$	vector of ones
$\mathbf{0}$	zero vector or matrix
$\mathbb{I}[\cdot]$	indicator function
F	number of activated sludge components ($F = 20$)
P	number of biological and chemical processes ($P = 28$)
K	dimension of the stoichiometric null space ($K = 5$)
q	number of independent stoichiometric invariants
N	training-set size
n	number of validation samples in a scored fold
M_{op}	number of operating variables ($M_{op} = 2$)
D	dimension of the second-order feature vector
i	sample index
j	component index
g	outer cross-validation fold index

m	surrogate model index
ν	stoichiometric (Petersen) matrix
A	stoichiometric invariant (conservation) operator
V_0	orthonormal basis of the stoichiometric null space
Z	orthonormal basis of $\text{null}(A)$
ρ	vector of biological and chemical process rates
D_i	hydraulic dilution rate of sample i
ξ	reaction progress variables
ω	dissolved-oxygen transfer coefficient
e_{SO}	dissolved-oxygen unit direction
c	activated sludge component vector
c_{in}	influent component vector
c_{out}	mechanistic effluent (target) component vector
b_i	sample conservation target, $Ac_{in,i}$
$\mathcal{F}_i$	feasible effluent set of sample i
I_{comp}	component-to-composite composition matrix
y	reported composite vector (COD, TN, TP, TSS)
σ_j	standard deviation of component j
u	operating-variable vector (HRT, aeration)
x	full surrogate input vector
f_m	trained surrogate model m
c_{raw}	raw surrogate component prediction
c_{cand}	conservative projection candidate
$\tilde{c}$	projected component prediction
ϵ_c	componentwise positivity floor
s	relative scaling vector
W_s	diagonal relative-scaling matrix
α	positivity backtracking factor
η	backtracking safety margin
δ	conservative candidate displacement
ϕ	second-order feature vector
Φ	training feature matrix

r	latent component-space forcing vector
B	regression driver coefficient matrix
Γ	cross-component coupling matrix
R	coupling residual matrix, $I_F - \Gamma$
W_u	first-order operating-effect block
W_{in}	first-order influent-load block
Θ_{uu}	operating-curvature block
Θ_{cc}	influent-curvature block
Θ_{uc}	operation-loading interaction block
λ_{inv}	invariant-penalty weight
λ_{sys}	coupled-system penalty weight
λ_B	driver ridge weight
λ_Γ	coupling ridge weight
$\hat{C}$	auxiliary nonnegative fitted-state matrix
$\mathcal{G}$	admissible coupling set
c_{aff}	affine invariant-projected state
$\hat{c}$	final checked deployed state
w_c	componentwise correction weights
$\mathcal{T}_{N,g}$	training partition at size N , fold g
ξ_{ij}	standardized component residual
R^2	coefficient of determination
π_{mc}	mass-conservation violation frequency
π_{nn}	nonnegativity violation frequency
v_{mc}	mass-conservation residual magnitude
v_{nn}	standardized negative-component magnitude
d_c	standardized projection displacement

Introduction

0.1 Review of Related Literature

0.1.1 Water Pollution, Biological Treatment, and Activated Sludge Modeling

Water pollution is a public health and environmental problem whose consequences depend on pollutant sources, exposure pathways, and the vulnerability of affected populations. Lin, Yang, and Xu (2022) connect polluted water to heterogeneous disease burdens, while Madhav et al. (2019) organize water pollutants by origin and effect across industrial, agricultural, domestic, and natural pathways. These reviews establish the wider environmental motivation for engineered treatment systems. Wastewater treatment is not only a matter of meeting a numerical effluent target. It is part of the larger effort to interrupt pollutant movement from human activity into aquatic ecosystems and exposed communities.

The activated sludge process remains a central biological route for municipal and industrial wastewater treatment because it brings microbial growth, organic carbon removal, nitrogen conversion, phosphorus behavior, oxygen transfer, and solids production into one engineered system. The process is complex because its treatment performance arises from interacting microbial and physicochemical pathways rather than from a single dominant reaction. Duan, Zhang, and Zheng (2022) show this clearly in their review of heterotrophic nitrifying bacteria, where nitrogen removal depends on microbial pathways whose importance changes with wastewater composition and operating conditions. Wentzel, Ekama, and Marais (1992) likewise present nitrification, denitrification, and biological excess phosphorus removal as coupled activated sludge processes. Their review helps explain why carbon, nitrogen, phosphorus, biomass, and solids must be treated as interdependent states when the process is modeled.

The Activated Sludge Model (ASM) family gives this interdependence a mechanistic language. Henze et al. (1999) introduced ASM2d as a model for biological phosphorus removal with simultaneous nitrification and denitrification, and Henze, Gujer, Mino, and van Loosdrecht (2015) consolidated the wider ASM family as a standard framework for design, operation, teaching, and research. The relevance of ASM2d is especially strong for work that must represent carbon, nitrogen, phosphorus, alkalinity,

and suspended solids together. Later developments also made explicit the need to represent nitrite-producing and nitrite-consuming steps when biological nutrient removal is studied in more detail. Guerrero, Guisasola, and Baeza (2011) used an ASM2d-based formulation with two-step nitrification and denitrification to examine competition between phosphorus accumulating organisms and denitrifiers. This component-based tradition frames activated sludge prediction as a state prediction problem rather than only an aggregate effluent estimation problem. The object of prediction is therefore not merely an aggregate effluent quality index but a complete activated sludge state from which reported composites can be derived.

0.1.2 Mechanistic Simulation and the Need for Surrogates

Mechanistic activated sludge models are valuable because they preserve the biological and chemical structure of the process, but that same structure makes repeated simulation demanding. Nonlinear kinetics, interacting operating variables, uncertain influent conditions, and calibration decisions all affect the behavior of a simulated treatment system. de Araujo, Gallani, Mulas, and Skogestad (2013) demonstrate this in their sensitivity analysis of optimal operation and controlled variable selection for an activated sludge process model. Their work shows that an operating decision depends on how model states respond across the process, not only on a single effluent quantity. Machado, Lafuente, and Baeza (2014) add a complementary calibration perspective by showing that ASM2d parameter identifiability is entangled with influent and operational uncertainty in full-scale plant data.

The burden becomes more visible when a model is embedded in design or optimization. Aboagye et al. (2021) formulate wastewater treatment network design as an optimization and sustainability assessment problem in which many configurations must be evaluated. Padron-Paez, Almaraz, and Roman-Martinez (2020) similarly place plant design within a multiobjective framework that balances treatment performance and sustainability criteria. Fang, Ni, Li, Sheng, and Yu (2011) show the same need in biological nutrient removal, where a simulation-based approach is used to optimize a full-scale treatment process. These applications require a model to be queried many times while operating points, treatment alternatives, and tradeoffs are explored. A detailed simulator can then become the limiting element in a workflow whose value depends on broad search.

Surrogate modeling addresses this pressure by replacing repeated high-cost simulations with faster statistical approximations over a defined operating domain. Bhosekar and Ierapetritou (2018) review surrogate modeling in process systems and emphasize that sampling, validation, feasibility analysis, and optimization must be treated as linked concerns. The wastewater literature shows the same practical motivation. Durkin, Otte, and Guo (2024) train surrogates on high-fidelity wastewater resource recovery simulations and use them inside process optimization. The significance of these

studies is not simply that surrogate models run quickly. Their deeper importance is that they allow mechanistic knowledge to be used repeatedly in design, screening, and decision support where a full simulator would be inconvenient or too slow.

0.1.3 Machine Learning Surrogates for Wastewater Prediction

Data driven wastewater modeling has moved from plant monitoring toward operational prediction and decision support. Newhart, Holloway, Hering, and Cath (2019) review performance analyses of wastewater treatment plants and note both the promise of process data and the difficulty of turning that data into reliable plant knowledge. More recent studies show the field moving toward actionable prediction. Mao et al. (2024) use artificial intelligence tools to balance effluent quality and aeration energy, S. J. Yang et al. (2025) use machine learning to predict waste activated sludge generation for operational efficiency, and Wang, Li, Bai, Yu, and Qu (2025) compare interpretable machine learning models with a mechanistic model for effluent nitrogen prediction. Xiao, Zhao, Zhang, and Yue (2026) further illustrates the growing interest in frameworks that combine mechanistic dynamics with data driven surrogates for activated sludge management. This movement also extends to process operation and control, as shown by data-mining work on aeration optimization, neural-network methods for setpoint and model predictive control, and reinforcement learning reviews for wastewater treatment process control (Asadi, Verma, Yang, & Mejabi, 2017; Croll, Ikuma, Ong, & Sarkar, 2023; Sadeghassadi, Macnab, Gopaluni, & Westwick, 2018).

The surrogate families used in this area differ in what they assume about the process. Partial least squares is a natural baseline because it creates a compact multivariate relationship among correlated variables. Wold, Sjostrom, and Eriksson (2001) describe partial least squares regression as a basic chemometric tool, and Khatri, Gupta, and Gupta (2020) review its use in water quality analysis. Its appeal is interpretability, especially when influent and operating variables are strongly correlated. Its limitation is that strongly nonlinear microbial behavior may not be captured without richer features or nonlinear extensions. Woo et al. (2009) show the value of kernel partial least squares for online estimation in an industrial cokes wastewater treatment plant, and C. Yang, Zhang, Huang, and Liu (2021) combine partial least squares with a relevance vector machine for adaptive effluent quality prediction. These studies show how linear interpretability and nonlinear predictive flexibility have both been pursued in wastewater modeling.

Tree ensembles, boosting methods, and neural networks offer different ways of handling nonlinear tabular relationships. Breiman (2001) introduced random forests as ensembles of randomized trees, while Chen and Guestrin (2016) developed XGBoost as a scalable boosted tree system for structured data. Neural approaches provide another form of flexibility. Hornik, Stinchcombe, and White (1989) established the approximation capacity of multilayer feedforward networks, and Arik and Pfister (2021)

introduced TabNet as a deep learning model for tabular data with attentive feature selection. In activated sludge surrogate modeling, these algorithm families are not interchangeable labels. They represent different compromises among interpretability, data efficiency, nonlinear response learning, computational cost, and stability outside the center of the training domain.

0.1.4 Physical Admissibility Beyond Predictive Accuracy

A central limitation of an unconstrained surrogate is that predictive accuracy does not by itself establish physical admissibility. A surrogate can be close to a mechanistic target under ordinary error metrics while still returning component concentrations that cannot be interpreted as a physically valid activated sludge state. The ASM tradition represents process transformations through stoichiometric relations among component states (Henze et al., 1999, 2015). Within a fixed control volume and model boundary, those relations imply material balances that the influent and effluent component states should respect. Concentrations must also remain nonnegative if they are to retain their physical meaning.

This distinction matters because reported composites such as chemical oxygen demand, total nitrogen, total phosphorus, and total suspended solids are summaries of an underlying component state. An aggregate value can look plausible while the component vector behind it contains negative concentrations or violates a declared material balance. For wastewater decision support, that mismatch is not a harmless numerical detail. It can distort interpretation, affect downstream optimization, and make a prediction less credible even when the aggregate error appears small.

The broader scientific machine learning literature reinforces this concern. Kashinath et al. (2021) review physics informed case studies in weather and climate modeling and show that scientific prediction must be judged against physical structure as well as empirical error. Hansen, Maddix, Alizadeh, Gupta, and Mahoney (2024) focus on conservation laws and argue that learned physical models may need explicit mechanisms to respect conserved quantities. P. M. Sturm and Wexler (2020) and P. M. Sturm and Wexler (2022) demonstrate the issue in photochemical modeling, where machine learning speedups must preserve mass and atom balances to remain credible. Shen et al. (2023) further connect data driven prediction with physical modeling by presenting differentiable modeling as a route for combining learned components with process-based structure. Kircher and Votsmeier (2025) make the paired requirement especially clear by showing that a conservative correction is incomplete when it creates negative species and that a positivity preserving adjustment is incomplete when it breaks atom balance. Although these studies arise outside activated sludge modeling, they identify the same conceptual problem that appears when a wastewater surrogate predicts a full component state.

0.1.5 Constraint Enforcement in Scientific Machine Learning

Scientific machine learning has developed several strategies for incorporating physical structure into data driven models. Physics informed approaches place governing relations, boundary conditions, or other physical penalties inside the training objective. Raissi, Perdikaris, and Karniadakis (2019) formalize physics informed neural networks for forward and inverse problems governed by differential equations, and Karniadakis et al. (2021) place such models within a broader physics informed learning program. Shen et al. (2023) similarly emphasize the value of connecting machine learning with process-based structure rather than treating the two as separate paradigms. These methods are important because they make physical knowledge visible during training. Their limitation for the present wastewater setting is that a penalty can encourage a physically reasonable prediction without guaranteeing that every deployed prediction satisfies the required balance and nonnegativity conditions.

Hard enforcement methods respond by building the physical requirement into the architecture, output representation, or correction step. Beucler et al. (2021) show that analytic constraints can be enforced in neural networks that emulate physical systems. Hansen et al. (2024) develop approaches for learning physical models that can respect conservation laws. P. M. Sturm and Wexler (2020) and P. M. Sturm and Wexler (2022) use conservative formulations for photochemical surrogates, while P. O. Sturm and Silva (2024) extend the conservation view through a species weighted correction for atmospheric composition. Valente, Dias, Guerra, and Ventura (2025) present output projection onto physical manifolds as a way to return model predictions to physically consistent sets. These works show that exact consistency can be imposed after, within, or around a learned model.

The activated sludge problem requires joint attention to conservation and nonnegativity. A conservation-only method can still produce negative concentrations, while a positivity-only transformation can disturb the case-specific balance implied by the mechanistic model. Kircher and Votsmeier (2025) address this combined requirement for mechanistic kinetics surrogates by embedding atom balance and positivity together. Their application domains differ from wastewater treatment, but the central physical demand is parallel. A corrected prediction must remain on the admissible balance relation while staying inside the domain of nonnegative concentrations.

0.1.6 Projection, Interpretability, and Dissertation Gap

Post prediction projection is attractive for activated sludge surrogates because it can be applied after a surrogate has already produced a component prediction. This makes the correction conceptually separate from the learning algorithm and allows the same physical check to be placed after linear, kernel, neighbor, ensemble, boosting, or neural predictors. The literature on conservation and output projection suggests that

such a step can return predictions to a physically consistent set (Kircher & Votsmeier, 2025; Valente et al., 2025). At the same time, projection is not a substitute for model selection. It can make a prediction admissible while changing ordinary error metrics in ways that depend on the surrogate, the component, and the operating regime. A physically enforced wastewater surrogate therefore still requires paired evaluation of predictive accuracy and constraint compliance.

Interpretability forms a second requirement for engineering usefulness. Rudin (2019) argues that high-stakes decisions should prefer models that are interpretable by design when such models can achieve sufficient performance. Wastewater treatment modeling is not identical to the decision settings emphasized in that argument, but the engineering concern is similar. If a surrogate is used for environmental decision support, its learned relationships should be inspectable whenever the modeling task permits. This concern links the older appeal of partial least squares, the current interest in interpretable tabular learning, and the need for structured physically admissible surrogates.

The resulting gap is specific. Activated sludge models provide the component-space structure needed to represent carbon, nitrogen, phosphorus, alkalinity, and solids transformations, but repeated mechanistic simulation can be too costly for broad exploration. Machine learning surrogates can accelerate prediction and have already shown value in wastewater treatment applications, but low error does not guarantee conservation, nonnegativity, or interpretability. Addressing this gap requires integrating surrogate comparison, post prediction physical enforcement, and interpretable constrained regression into one activated sludge modeling program. The aim is a surrogate framework that is fast enough for repeated use, accurate enough for engineering analysis, physically admissible at the component level, and transparent enough to support engineering judgment.

0.2 Statement of the Problem

Activated sludge models provide a rigorous mechanistic basis for representing biological wastewater treatment, but their use becomes burdensome when many operating or influent conditions must be evaluated. At the same time, machine learning surrogates can accelerate prediction but may return component states that are accurate under conventional error metrics while remaining physically inadmissible. The central problem is therefore to determine how machine learning surrogates of the activated sludge process can be developed and evaluated so that their predictions remain useful not only as fast approximations, but also as physically meaningful component-space states.

The problem begins with the predictive modeling setting itself. Wastewater treatment applications increasingly use data driven models for effluent quality prediction,

operational support, and process optimization (Mao et al., 2024; Newhart et al., 2019; Wang et al., 2025; S. J. Yang et al., 2025). However, the activated sludge process is not a simple single-output regression problem. The state described by ASM2d-based models contains interacting carbon, nitrogen, phosphorus, alkalinity, and solids components (Guerrero et al., 2011; Henze et al., 1999, 2015). A surrogate that predicts only reported aggregate quantities may be useful for limited monitoring tasks, but it does not directly preserve the component-space information needed to examine conservation, nonnegativity, and downstream composition calculations. A rigorous assessment must therefore establish how a range of statistical surrogates behave when they are asked to predict the complete activated sludge component state rather than only an aggregate water quality summary.

The second part of the problem concerns physical admissibility. Conventional surrogate assessment emphasizes prediction error, but a low error value does not guarantee that an activated sludge prediction satisfies the material balance structure implied by the adopted mechanistic model. It also does not guarantee that every component concentration remains nonnegative. This creates a gap between numerical accuracy and engineering credibility. Studies in scientific machine learning show that conservation and positivity often require explicit treatment rather than passive expectation from data fitting alone (Beucler et al., 2021; Hansen et al., 2024; Kircher & Votsmeier, 2025; P. M. Sturm & Wexler, 2020, 2022). It is therefore necessary to determine whether post prediction enforcement can consistently return activated sludge surrogate outputs to the physically admissible component space, and whether that enforcement improves, worsens, or leaves unchanged the usual measures of predictive performance.

The third part of the problem concerns interpretability. A physically corrected prediction can be useful for screening and computation, but correction alone does not necessarily make the learned relationship between operating conditions, influent composition, and effluent state understandable. For engineering decision support, the structure of the surrogate matters because the prediction may inform operating judgment, design exploration, or later optimization. The research therefore examines whether an interpretable surrogate can be structured so that it retains transparent relationships among operating variables, influent loads, nonlinear effects, and component coupling while still returning physically admissible predictions. In this sense, the research problem is not merely to reduce prediction error. It is to integrate accuracy, computational speed, physical admissibility, and interpretability within one activated sludge surrogate modeling framework.

0.3 Significance of the Study

This study is significant because it addresses a practical weakness in the use of machine learning for biological wastewater treatment modeling. Wastewater treatment

systems protect receiving waters and public health by reducing pollutant loads before discharge, and their operation depends on reliable understanding of process behavior (Lin et al., 2022; Madhav et al., 2019). As treatment modeling moves toward data driven prediction and optimization, the credibility of a surrogate model cannot rest only on speed or ordinary error metrics (Asadi et al., 2017; Croll et al., 2023; Fang et al., 2011; Sadeghassadi et al., 2018). A prediction that violates mass conservation or contains negative component concentrations may be numerically close to a target but still unsuitable for engineering interpretation (Hansen et al., 2024; Kircher & Votsmeier, 2025; Shen et al., 2023). By placing physical admissibility at the same level of importance as accuracy, the study strengthens the basis on which surrogate predictions can be used in wastewater treatment analysis.

The study contributes to activated sludge modeling by keeping the complete component state at the center of the surrogate problem. The ASM family was developed to represent interacting biological and chemical processes, not only final aggregate effluent indicators (Henze et al., 1999, 2015). A component-space surrogate retains that mechanistic orientation while reducing the cost of repeated evaluation. This matters for sensitivity analysis, design screening, operating exploration, and optimization workflows where many simulations may be needed (Aboagye et al., 2021; de Araujo et al., 2013; Durkin et al., 2024; Padron-Paez et al., 2020). The work therefore helps connect the mechanistic detail of activated sludge models with the computational convenience expected from modern surrogate modeling.

The study also offers methodological value for wastewater applications of machine learning. It separates ordinary predictive accuracy from physical constraint compliance and treats both as necessary diagnostics for complete component predictions. This distinction is important because different surrogate families may show different error behavior, different tendencies toward negative values, and different responses to physical projection. The resulting evaluation logic can help future wastewater surrogate studies report not only whether a model predicts well, but also whether its predictions remain admissible under the declared process boundary and component basis.

Finally, the study is significant for engineering decision support because it addresses interpretability alongside feasibility. Interpretable models can be preferable in high-consequence settings when their performance is sufficient, since they allow the structure of the learned relationship to be inspected rather than treated only as a black-box output (Rudin, 2019). In the activated sludge setting, interpretability supports the examination of how operating conditions and influent loads are associated with the predicted effluent state. A physically admissible and interpretable surrogate can therefore serve as a bridge between mechanistic simulation and practical decision workflows, providing predictions that are faster than repeated simulation, more physically defensible than unconstrained regression outputs, and more inspectable than purely black-box approximations.

0.4 Research Framework

The research framework is organized as a Theory of Change. In evaluation and intervention research, a Theory of Change makes explicit how a set of activities is expected to produce outputs, outcomes, and longer-term value under stated assumptions. Mayne (2015) emphasizes that useful Theory of Change models clarify the causal logic of a program rather than merely listing its components. De Silva et al. (2014) and Breuer, Lee, De Silva, and Lund (2016) similarly show that Theory of Change reasoning can strengthen the design and evaluation of complex interventions by connecting context, mechanisms, evidence, and intended change. In this research, the intervention is methodological rather than institutional. The intended change is a more credible way of using machine learning surrogates for activated sludge modeling, where computational speed is joined with component-level physical admissibility and interpretability.

The starting condition is the tension identified in the review of related literature. Mechanistic activated sludge models encode the biological and chemical structure of wastewater treatment, but repeated simulation can be too costly for design exploration, screening, and optimization. Machine learning surrogates address the computational burden, but unconstrained predictions may violate conservation or nonnegativity even when ordinary error metrics are favorable (Hansen et al., 2024; Kircher & Votsmeier, 2025; Shen et al., 2023). The Theory of Change therefore begins from the premise that wastewater surrogate modeling should be evaluated not only by closeness to simulated targets, but also by whether the returned component state remains physically meaningful under the adopted ASM boundary.

The framework rests on the assumption that complete component-state prediction is more appropriate than direct aggregate-only prediction when the purpose is to preserve activated sludge process structure. It also assumes that physical admissibility must be checked after prediction because it is not guaranteed by low error alone, and that engineering usefulness is strengthened when the learned relationships among operating conditions, influent loads, and effluent components can be inspected. These assumptions shape three connected research activities. The first activity establishes the surrogate prediction setting and compares statistical learners for complete activated sludge component states. The second activity evaluates post prediction enforcement of mass conservation and nonnegativity across surrogate families and data regimes. The third activity develops an interpretable physically admissible surrogate that integrates structured regression with checked deployment.

The pathway in Fig. 1 makes the research logic explicit. The context defines the practical and scientific need. The assumptions identify what must be true for the research program to matter in activated sludge modeling. The activities correspond to the empirical and methodological tasks, and the outputs are the concrete products of those tasks. The intended outcome is not the replacement of mechanistic activated

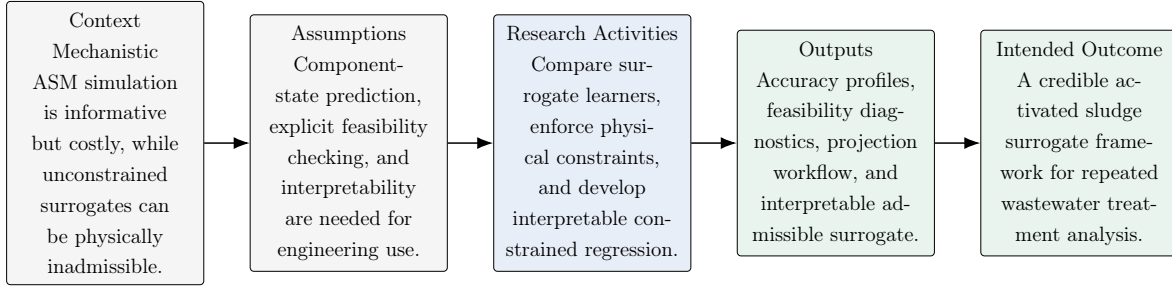


Figure 1: Theory of Change for the research framework.

sludge models, but a surrogate modeling framework that can support repeated wastewater treatment analysis while preserving the physical meaning of predicted component states.

The Theory of Change also clarifies how evidence is interpreted across the research. If a surrogate gives low prediction error but violates conservation or nonnegativity, the framework treats that result as incomplete rather than successful. If projection restores physical admissibility but changes prediction error, the framework treats the change as an empirical tradeoff to be reported, not as an automatic failure or improvement. If an interpretable constrained surrogate sacrifices some fixed-split accuracy while improving physical reliability and structural transparency, the framework treats that result as part of a broader engineering decision space. The framework therefore supports a balanced interpretation of speed, accuracy, physical admissibility, and interpretability as connected outcomes of the study rather than as isolated metrics.

0.5 Scope and Limitations

This study is scoped to machine learning surrogates of the activated sludge process under an ASM2d-based component representation. The mechanistic foundation is the ASM family and its extension to biological phosphorus removal and two-step nitrogen conversion (Guerrero et al., 2011; Henze et al., 1999, 2015). The surrogate task is defined at the level of activated sludge component states rather than only at the level of reported effluent composites. Chemical oxygen demand, total nitrogen, total phosphorus, and total suspended solids remain important for interpretation, but they are treated as derived reporting quantities rather than as the only prediction targets.

The empirical scope is limited to simulated steady-state reactor data. The study does not claim direct field validation on full-scale wastewater treatment plant measurements, and it does not attempt to represent every operational disturbance, sensor error, seasonal shift, or unmodeled biological pathway that may occur in practice. The simulated setting is nevertheless appropriate for the present purpose because the study asks how surrogate predictions behave when the mechanistic reference state, component basis, and physical constraint structure are known. This allows prediction error,

mass conservation, nonnegativity, and correction behavior to be examined without confounding them with measurement noise or uncertain plant instrumentation.

The modeling scope includes a comparative set of statistical surrogates and an interpretable constrained surrogate developed in the study. The comparison is intended to represent major surrogate families used for structured process data, including linear latent-variable methods, neighbor and kernel methods, tree ensembles, boosting methods, and neural networks. It is not an exhaustive survey of every possible machine learning architecture, every hyperparameter strategy, or every hybrid mechanistic data driven formulation. The findings should therefore be interpreted as evidence within the evaluated surrogate set and benchmark protocol, not as a universal ranking of all possible wastewater treatment predictors.

The physical guarantees considered here are deliberately specific. Conservation refers to the stoichiometric invariant structure implied by the adopted activated sludge model and system boundary, while nonnegativity refers to the component concentrations returned by the surrogate. These guarantees do not imply complete kinetic realism, thermodynamic validity, microbial ecological correctness, or plant-wide mass closure outside the declared boundary. A prediction can be physically admissible in the constrained component-space sense and still require further scrutiny before being used for detailed plant design, regulatory reporting, or closed-loop control.

The evaluation also has a bounded generalization claim. It examines performance within the sampled operating and influent domain and under controlled excursions beyond that domain, but it does not prove reliability for arbitrary influent compositions, reactor configurations, treatment trains, climates, or operational policies. The beyond-domain evaluations are useful because they reveal how prediction error and physical enforcement behave when inputs move outside the training envelope. They should not be read as proof that the surrogates will remain accurate under every future plant condition.

Finally, the interpretability claim is limited to the structure of the surrogate and the inspectability of its learned relationships. A transparent coefficient block or component coupling pattern can support engineering interpretation, but it should not be treated as a causal statement about the activated sludge process without additional experimental or mechanistic analysis. Within these boundaries, the study contributes a framework for evaluating and constructing activated sludge surrogates whose predictions are fast, physically admissible under the stated constraints, and more interpretable than unconstrained black-box outputs.

Chapter 1

Machine Learning Surrogates of the Activated Sludge Process

1.1 Rationale

Activated sludge processes remain central to biological wastewater treatment because they link organic-matter removal, nutrient transformation, biomass growth, solids handling, and aeration control in one coupled process. Mechanistic activated sludge models provide a scientific language for representing these interactions, but their practical use can become difficult when many operating conditions or influent scenarios must be examined (Guerrero et al., 2011; Henze et al., 1999, 2015). Scenario screening, operational assessment, and process exploration therefore benefit from predictive tools that can approximate mechanistic behavior more quickly while retaining enough fidelity to support engineering interpretation.

Machine learning surrogates are attractive for this purpose because they can learn nonlinear relationships from structured process data and return predictions at much lower computational cost than repeated mechanistic simulation. Wastewater treatment studies increasingly use data driven models for effluent prediction, operational decision support, energy assessment, and plant-performance analysis (Asadi et al., 2017; Croll et al., 2023; Mao et al., 2024; Newhart et al., 2019; Sadeghassadi et al., 2018; Wang et al., 2025; S. J. Yang et al., 2025). This growing literature shows that machine learning can be useful in wastewater applications, but it does not by itself establish which modern tabular learners are most effective as raw surrogates of an activated sludge simulator.

The gap is partly methodological and partly substantive. Many wastewater prediction studies emphasize plant monitoring data, selected effluent indicators, or a small number of model families. These studies are valuable, but they make it difficult to separate the learnability of the activated sludge response from site-specific noise, limited target definitions, or algorithm-specific choices. A controlled surrogate benchmark is needed to show whether a broad set of state of the art tabular models can learn

the nonlinear response of an activated sludge process from paired inputs and outputs before any post prediction correction or additional physical enforcement is considered (Borisov et al., 2024; Shwartz-Ziv & Armon, 2022).

Such a baseline matters because raw predictive accuracy is the first requirement of a useful surrogate. If the unadjusted model output cannot recover the behavior of the mechanistic simulator, then later use in screening, comparison, or decision support will rest on a weak empirical foundation. If raw predictions are accurate, however, they establish that contemporary tabular models can capture important nonlinear structure in the activated sludge response. This provides a focused basis for comparing model families, data efficiency, and robustness under changes in operating conditions without conflating raw learning ability with downstream correction procedures.

1.2 Method

The analysis used a fixed dataset generated from the mechanistic activated sludge simulator under the protocol described below. The same in-distribution samples, beyond-domain samples, train-validation splits, selected model configurations, preprocessing rules, and raw surrogate predictions were used throughout the evaluation. No additional simulator runs, fold reassignments, hyperparameter searches, or alternative preprocessing rules were introduced after the dataset and model protocol were finalized.

1.2.1 Data Source and Prediction Scope

The data were generated from a standalone, steady-state continuous stirred tank reactor represented by the Activated Sludge Model No. 2d with two-step nitrification (ASM2d-TSN). The process description retained the 20 activated sludge components and 28 biological and chemical processes associated with hydrolysis, heterotrophic and phosphate-accumulating-organism metabolism, two-step nitrification and denitrification, lysis, and precipitation-redissolution. The model basis followed the Activated Sludge Model family documentation of Henze et al. (2015), while the two-step nitrification formulation, stoichiometry, and rate coefficients followed Guerrero et al. (2011). This mechanistic simulator supplied the reference input-output pairs used to train and evaluate the statistical surrogates.

Only raw surrogate predictions were analyzed. Scoring used the unadjusted effluent component vectors returned by the fitted surrogates. Predictions were not clipped, reconciled, projected, or otherwise corrected after inference. This endpoint measures whether tabular surrogates can approximate the nonlinear activated sludge mapping from the same inputs and outputs used by the mechanistic benchmark.

1.2.2 Mechanistic Benchmark and Prediction Task

The in-distribution design sampled hydraulic retention time, aeration, and 20 influent component concentrations, giving 22 continuous input variables. A randomized, scrambled strength-one Latin hypercube was generated with root seed 42 and linearly scaled to the prespecified operating and influent bounds (McKay, Beckman, & Conover, 1979). Each sampled input was passed to the steady-state ASM2d-TSN simulator. The steady-state equations were solved by nonlinear least squares under componentwise state bounds from a lower numerical floor to an upper bound of 1500 in each component’s native unit. Solver cost-change, step-size, and gradient tolerances were set to 10^{-9} , and each nonlinear solve was limited to 1200 function evaluations. Two physically informed initial guesses were attempted before a 20-day stiff backward differentiation formula dynamic relaxation was used as a fallback initializer. Accepted states were required to converge with a maximum absolute balance residual no greater than 10^{-9} . The in-distribution dataset retained 10,000 accepted steady states.

The retained steady states were converted into a supervised tabular regression problem. The input vector for each row consisted of hydraulic retention time, aeration, and the complete influent component vector. The target was the corresponding 20-component effluent state. Reported composite quantities such as chemical oxygen demand, total nitrogen, total phosphorus, and total suspended solids were computed after component prediction using a fixed component-to-composite composition map. These composites were therefore interpretive summaries of the predicted component state rather than separately fitted response variables.

Two beyond-domain validation sets were generated by the same simulator and acceptance rules. These sets evaluated raw surrogate behavior outside the original sampling envelope without introducing post prediction correction. Each beyond-domain regime contains 600 accepted steady states. The mild regime perturbed selected inputs 10% to 25% beyond their original upper bounds, while the severe regime perturbed them 35% to 60% beyond those bounds. The perturbed features were hydraulic retention time, aeration, influent ammonium nitrogen, and influent slowly biodegradable substrate. Candidate generation used a mixture dominated by one-feature excursions with a smaller share of two-feature excursions, while all other inputs remained inside their in-distribution ranges.

1.2.3 Surrogate Models and Preprocessing

Thirteen statistical surrogates were evaluated from their raw predictions. The model set was chosen to represent the main families commonly used for structured tabular regression, including boosted trees, randomized tree ensembles, kernel and neighbor methods, latent-variable regression, sparse multi-task linear regression, and neural-network surrogates. This selection allowed flexible nonlinear learners to be com-

pared with simpler or more inspectable alternatives while holding the simulation data, folds, and scoring rules fixed. The model families draw on established surrogate and tabular-learning foundations, including random forests, gradient boosting, partial least squares, and attentive tabular neural networks (Arik & Pfister, 2021; Borisov et al., 2024; Breiman, 2001; Chen & Guestrin, 2016; Wold et al., 2001).

Table 1.1: Surrogate groups and preprocessing used for raw-prediction evaluation.

Surrogate group	Models	Output handling	Preprocessing before raw scoring
Boosting	XGBoost, LightGBM, CatBoost, AdaBoost	One regressor per effluent component	XGBoost, LightGBM, and CatBoost used physical input and target units; AdaBoost used target z-scores only
Randomized tree ensembles	Random Forest, Extra Trees	Native multi-output regression	Random Forest used physical input and target units; Extra Trees used target z-scores only
Kernel and neighbor methods	Support vector regression (SVR), k-nearest neighbors (k-NN)	SVR used one regressor per component; k-NN returned all components jointly	Input and target z-scores fitted within the relevant training partition
Latent-variable and sparse multi-task models	Partial least squares (PLS), Multi-task Elastic Net, Multi-task Lasso	Joint 20-component regression	Input and target z-scores fitted within the relevant training partition
Neural-network surrogates	Multilayer perceptron (MLP), TabNet	Joint 20-component regression	Input and target z-scores fitted within the relevant training partition

All transformations were fitted only on the relevant inner-training, outer-training, or full in-distribution training partition. The same fitted transformations were then applied to the corresponding validation or beyond-domain inputs. When a model was trained on target z-scores, its predictions were transformed back to physical component units before any score was computed. This training-partition scaling rule is important because feature scaling can materially affect regression behavior and model comparison (Pinheiro et al., 2025). The workflow did not use missing-value imputation, response-dependent outlier removal, or target filtering. Consequently, the reported raw-prediction results reflect each surrogate’s learned mapping under a consistent and leakage-controlled preprocessing scheme.

1.2.4 Training, Tuning, and Validation Protocol

In-distribution performance was evaluated with nested sample sizes of 500, 1450, 2400, 3350, 4300, 5250, 6200, 7150, 8100, 9050, and 10,000 rows. A seed-42 permutation established the nested row order and persistent master fold assignments before any subset was formed. At every sample size, five-fold cross-validation used the same

validation folds for every surrogate. Each fold therefore trained on four fifths of the selected subset and scored the remaining fifth. The common folds made the raw surrogate comparisons depend on model behavior rather than on different validation samples.

Hyperparameters were selected without access to the scored outer fold. Within each largest-scale outer training partition, a four-fold inner search minimized component-standardized mean squared error. The selected setting for a given outer fold was then held fixed when the same surrogate was evaluated on smaller nested subsets belonging to that fold. This design made the learning curves a controlled sample-size analysis rather than a sequence of separately optimized small-data studies. Hyperparameter optimization used seeded Optuna studies with a tree-structured Parzen estimator and no pruning (Akiba, Sano, Yanase, Ohta, & Koyama, 2019). Each surrogate had five outer-fold studies at the largest in-distribution scale and one additional full in-distribution study for the beyond-domain refit, with 100 trials per study.

After the in-distribution cross-validation analysis, one final setting per surrogate was selected using only the 10,000 in-distribution states. Each surrogate and its required preprocessing were then refitted on the full in-distribution pool before raw predictions were made for the mild and severe beyond-domain sets. No beyond-domain response was used for refitting, scaling, calibration, or hyperparameter selection.

1.2.5 Raw Prediction Evaluation

The primary outcome was accuracy of the 20-component effluent prediction. Component errors were scored in standardized form so that components expressed in different physical units could contribute to one aggregate assessment. The raw-prediction metrics were normalized mean squared error, normalized root mean squared error (nRMSE), normalized mean absolute error, and the macro-average coefficient of determination across effluent components. The training reference set associated with the scored fold supplied the component standard deviations used for normalization. Fold-level results were summarized descriptively as means and sample standard deviations across the five outer folds. No hypothesis tests were applied.

Learning curves were constructed from the same fold summaries across the nested sample sizes. These curves showed how quickly each surrogate converted additional mechanistic simulations into lower raw prediction error, which is central to judging whether the activated sludge mapping was learnable from tabular input-output data (Viering & Loog, 2023). Component-specific physical-unit errors were retained for diagnosing which effluent components were easier or harder to approximate. Composite chemical oxygen demand, total nitrogen, total phosphorus, and total suspended solids metrics were treated as secondary summaries computed from the raw component predictions.

Beyond-domain evaluation used the full-data refit of each surrogate and the two separately simulated validation regimes. The same raw component and composite metrics

were computed for the mild and severe datasets, using the in-distribution training pool as the normalization reference. These beyond-domain scores were interpreted descriptively over the 600 retained cases in each regime. They were not averaged across folds because the beyond-domain predictions came from one final refit per surrogate. Elapsed-time measurements were used only to characterize the computational cost of fitting, preprocessing, and raw inference.

A surrogate was judged successful to the extent that its unadjusted predictions recovered the simulated effluent component state, improved with additional training data, and retained useful accuracy under controlled beyond-domain shifts. The endpoint was predictive approximation of the mechanistic simulator’s component outputs, with water-quality composites used only as downstream summaries of those component predictions.

1.3 Results

The raw surrogate benchmark showed that the activated sludge response was learnable from paired simulated inputs and outputs, but the degree of learning depended strongly on the model family and the number of training cases. Table 1.2 summarizes the main raw-prediction results. At the smallest training size, the strongest performers were boosted trees. XGBoost and LightGBM had nearly identical mean nRMSE values of 0.435 and 0.439, respectively, while CatBoost followed at 0.493. The MLP was competitive but not dominant at this low-data setting, with a mean nRMSE of 0.524. TabNet had the weakest small-sample performance, with a mean nRMSE of 0.938, showing that its deep tabular architecture required more data before it became competitive.

Table 1.2: Raw surrogate performance within and beyond the sampled domain. In-distribution values are five-fold means. Beyond-domain values are computed from the full in-distribution refit in each regime.

Surrogate	Family	nRMSE at 500	nRMSE at 10,000	Macro R ² at 10,000	Mild nRMSE	Severe nRMSE	Raw latency ms/sample
MLP	Neural networks	0.524	0.150	0.977	0.255	0.531	0.005
CatBoost	Boosting	0.493	0.230	0.947	0.393	0.836	0.068
LightGBM	Boosting	0.439	0.249	0.938	0.419	0.848	0.097
XGBoost	Boosting	0.435	0.254	0.935	0.424	0.861	0.060
TabNet	Neural networks	0.938	0.272	0.926	0.336	0.563	0.023
SVR	Kernel	0.511	0.364	0.868	0.555	0.989	1.874
AdaBoost	Boosting	0.550	0.525	0.724	0.692	1.036	1.031
Multi-task Lasso	Interpretable models	0.551	0.536	0.713	0.751	1.146	0.001
Multi-task Elastic Net	Interpretable models	0.552	0.536	0.713	0.750	1.146	0.001
Extra Trees	Randomized tree ensembles	0.711	0.563	0.683	0.717	1.084	0.146
k-NN	Instance based	0.747	0.607	0.632	0.760	1.113	0.157
Random Forest	Randomized tree ensembles	0.815	0.718	0.484	0.908	1.242	0.138
PLS	Interpretable models	0.744	0.721	0.480	0.908	1.267	0.001

Increasing the training set changed the ranking. The MLP improved from 0.524 at 500 cases to 0.150 at 10,000 cases, corresponding to a 71.3% reduction in mean nRMSE.

TabNet showed a similarly large relative reduction, falling from 0.938 to 0.272, but its final error remained higher than the MLP and the leading boosted trees. CatBoost improved from 0.493 to 0.230, while LightGBM and XGBoost improved from 0.439 to 0.249 and from 0.435 to 0.254, respectively. These results indicate that boosted trees were more data efficient at the smallest sample size, whereas the MLP used the larger training set most effectively. At the full in-distribution scale, the five best raw surrogates were the MLP, CatBoost, LightGBM, XGBoost, and TabNet. Their macro-average coefficients of determination ranged from 0.926 to 0.977, indicating that the best raw models captured most of the component-level variation in the simulator outputs. Figure 1.1 traces these trajectories across the full range of training-set sizes and shows the crossover in which the MLP overtook the boosted trees as more simulated data became available.

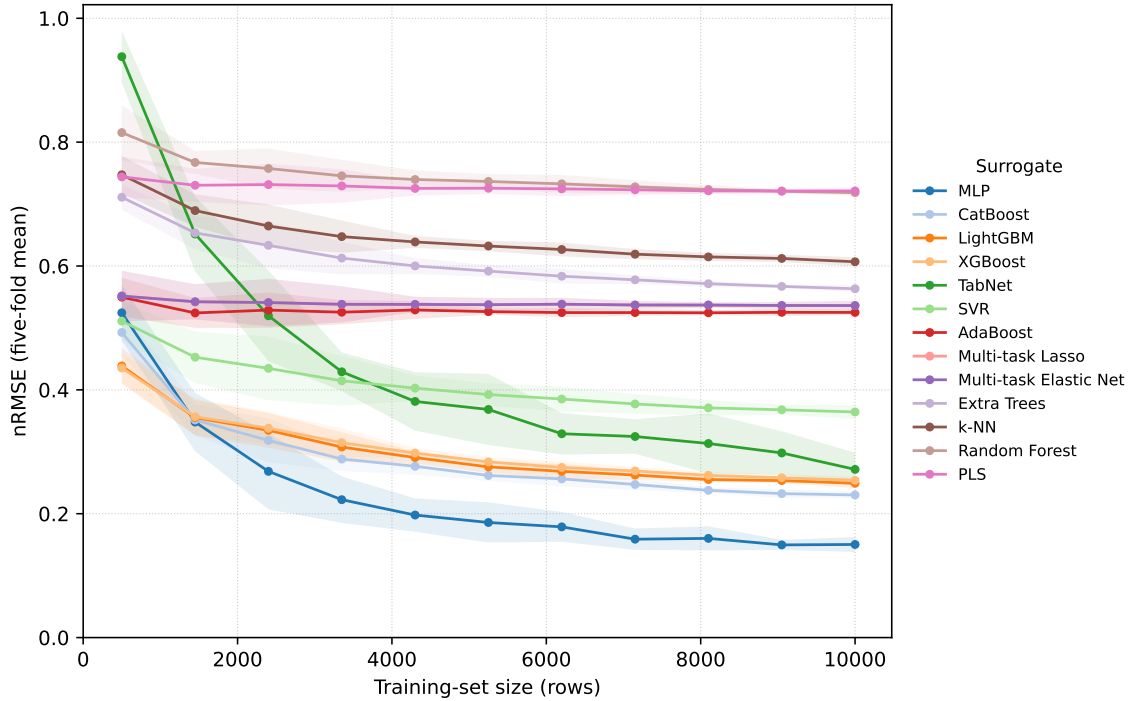


Figure 1.1: In-distribution learning curves for raw predictions. Lines show five-fold mean nRMSE and shaded regions show $\pm$ one sample SD across folds.

The remaining surrogate families learned the mapping less effectively. SVR reached a mean nRMSE of 0.364 at the largest sample size and remained clearly below the leading neural and boosted-tree models. AdaBoost, Multi-task Lasso, Multi-task Elastic Net, Extra Trees, k-NN, Random Forest, and PLS all retained full-size nRMSE values above 0.52. The multi-task linear models and PLS were fast and structurally simpler, but their raw errors were much higher than those of the most flexible nonlinear surrogates. Random Forest and Extra Trees also lagged behind the boosted trees, suggesting that averaging randomized trees without the same boosting mechanism was less effective for this multicomponent activated sludge response.

The component-level results showed that prediction difficulty was not uniform across the effluent state. Table 1.3 summarizes the components with the highest and lowest mean nRMSE at the largest in-distribution sample size, averaged across the 13 raw surrogates after first averaging over folds. Readily biodegradable substrate, nitrite nitrogen, nitrogen gas, metal hydroxide, and the polyhydroxyalkanoate storage component were the hardest components on this aggregate basis. Alkalinity, soluble inert organic matter, particulate inert organic matter, phosphate-accumulating organisms, and slowly biodegradable substrate were easier. The MLP reduced errors substantially across the component set, but even for the MLP the readily biodegradable substrate remained the hardest component, with a component nRMSE of 0.453.

Table 1.3: Component-level raw prediction difficulty at the largest in-distribution sample size, averaged across surrogate families.

Higher-error component	Mean nRMSE	Mean R ²	Lower-error component	Mean nRMSE	Mean R ²
S_F	0.734	0.422	S_{ALK}	0.157	0.942
S_{NO2}	0.644	0.525	S_I	0.162	0.939
S_{N2}	0.550	0.653	X_I	0.183	0.929
X_{MeOH}	0.523	0.678	X_{PAO}	0.263	0.896
X_{PHA}	0.493	0.722	X_S	0.329	0.847

The derived water-quality composites confirmed that low component error translated into accurate aggregate reporting for the leading raw models. For the MLP at the largest in-distribution size, the mean composite RMSE values were 5.35 g COD/m³ for chemical oxygen demand, 1.17 g N/m³ for total nitrogen, 0.51 g P/m³ for total phosphorus, and 4.76 g TSS/m³ for total suspended solids. The corresponding coefficients of determination ranged from 0.991 to 0.995. CatBoost produced similarly strong aggregate results for chemical oxygen demand and total suspended solids, but its total nitrogen and total phosphorus errors were higher than those of the MLP. These composite results are consistent with the component-level ranking and show that the strongest raw surrogate did not merely fit isolated components; it also preserved the aggregate water-quality summaries derived from the component predictions.

Beyond-domain evaluation produced the expected increase in error. Under the mild regime, the MLP remained the best raw surrogate with an nRMSE of 0.255 and a macro-average coefficient of determination of 0.964. TabNet ranked second with an nRMSE of 0.336, followed by CatBoost, LightGBM, and XGBoost. Under the severe regime, all models worsened, but the leading order remained stable. The MLP had an nRMSE of 0.531, TabNet had an nRMSE of 0.563, and the three boosted-tree models ranged from 0.836 to 0.861. The severe regime therefore remained predictable for the strongest neural surrogates but was substantially harder for boosted trees and much harder for the linear, neighbor, and randomized-ensemble baselines.

Raw inference was fast for every surrogate relative to mechanistic simulation. The multi-task linear models and PLS returned predictions in about 0.001 ms per sample,

but their accuracy was limited. The MLP combined the lowest full-size error with a mean raw inference latency of 0.005 ms per sample. TabNet, XGBoost, CatBoost, and LightGBM also remained below 0.1 ms per sample. SVR and AdaBoost were slower, at 1.874 and 1.031 ms per sample, respectively, while still producing higher errors than the leading neural and boosted-tree models. The timing results therefore reinforce the practical value of raw surrogates because the best-performing models were not only accurate, but also fast enough for repeated prediction over many candidate operating conditions.

1.4 Synthesis

The results support the central premise that modern tabular machine learning surrogates can learn a multicomponent activated sludge simulator response from paired input-output data. The strongest evidence is the performance of the MLP at the largest sample size. Its low nRMSE, high macro-average coefficient of determination, strong composite metrics, and very low raw inference latency show that the simulator response contains learnable nonlinear structure that can be captured by a flexible data driven model. This finding is consistent with the broad role of neural networks as nonlinear approximators and with their use in chemical and wastewater process modeling (Curteanu & Cartwright, 2011; Hornik et al., 1989; Sadeghassadi et al., 2018).

The comparison also shows why a single surrogate family should not be assumed to dominate under every data condition. Boosted trees were the strongest low-data learners, with XGBoost and LightGBM leading at 500 training cases. Their advantage at small sample size is consistent with the wider tabular-learning literature, where tree-based methods often remain strong baselines for structured datasets (Borisov et al., 2024; Shwartz-Ziv & Armon, 2022). As more simulated data became available, the MLP surpassed the boosted trees and retained the best beyond-domain accuracy. TabNet followed a related but less favorable trajectory, starting from poor small-data performance and improving sharply with more data. The learning curves therefore show that sample size changes the preferred surrogate family, which is why surrogate selection should be evaluated across more than one training-set size (Viering & Loog, 2023).

The weaker performance of PLS, the multi-task linear models, k-NN, Random Forest, and Extra Trees clarifies the character of the activated sludge prediction problem. Linear and latent-variable structure alone was not enough to recover the full component response with high accuracy, even though those models were fast and more inspectable. Neighbor averaging and randomized tree averaging also underperformed relative to boosted trees and neural networks. The response surface therefore appears to require flexible nonlinear representation, but not all nonlinear learners benefited equally from the available data. The strongest raw surrogates were those that com-

bined nonlinear flexibility with effective use of the tabular training structure.

The component-level findings add an important qualification. Aggregate accuracy did not mean that all effluent components were equally easy to predict. Several nitrogen-linked and substrate-related components remained more difficult across surrogate families, while alkalinity and inert components were comparatively easier. This pattern is chemically plausible because activated sludge nutrient and substrate transformations are coupled through nonlinear microbial pathways rather than through independent one-component responses (Guerrero et al., 2011; Henze et al., 2015). Consequently, model assessment should retain component-level diagnostics even when composite water-quality metrics are strong.

The beyond-domain results show that raw surrogate learning remained useful outside the sampled operating envelope but became less reliable as the shift intensified. The MLP and TabNet retained the best severe-regime accuracy, whereas boosted trees experienced a larger increase in error and the weaker baselines degraded substantially. This behavior is consistent with the general expectation that data driven models are less secure when evaluated outside the support of their training data, even when they interpolate well within the sampled domain (Karniadakis et al., 2021; Shen et al., 2023; Valente et al., 2025). The raw surrogates are therefore appropriate for rapid screening within and near the simulated domain, but beyond-domain use should be interpreted as extrapolative prediction rather than as guaranteed process fidelity.

Taken together, the results establish a raw-prediction baseline for activated sludge surrogates. Flexible tabular models, especially the MLP and the leading boosted trees, can approximate the mechanistic component response with useful accuracy and very low inference cost. The evidence also shows that raw accuracy depends on model family, training-set size, component identity, and degree of domain shift. The most defensible interpretation is therefore neither that any machine learning model is sufficient nor that mechanistic simulation can be casually replaced. Rather, the findings show that carefully validated raw surrogates can serve as fast approximations of activated sludge simulator behavior, provided that their component-level and beyond-domain limitations remain visible in the evaluation.

Chapter 2

Mass Conservation and Nonnegativity Enforcement on Machine Learning Surrogates of the Activated Sludge Process

2.1 Rationale

Once a surrogate has demonstrated that the activated sludge response is learnable from paired simulation data, a second credibility problem remains. The prediction is not only a numerical estimate of several effluent quantities; it is a component vector that should still be interpretable as an activated sludge state. The Activated Sludge Model family represents carbon, nitrogen, phosphorus, alkalinity, and solids through coupled stoichiometric relations, and those relations make conservation and nonnegativity part of the meaning of the predicted state (Guerrero et al., 2011; Henze et al., 2015). A surrogate that is close to the mechanistic target under ordinary error metrics can still return an output that violates those requirements, leaving a gap between statistical accuracy and physical admissibility (Hansen et al., 2024; Kircher & Votsmeier, 2025).

This gap is important because complete component predictions are often the basis for downstream water-quality interpretation. Composite quantities such as chemical oxygen demand, total nitrogen, total phosphorus, and total suspended solids summarize the component state, but they do not by themselves prove that the underlying vector satisfies the process structure from which those summaries are derived. A component vector can therefore look useful at the aggregate level while containing a negative concentration or violating a declared conservation relation. In wastewater treatment analysis, such an output is difficult to treat as a credible process state, particularly when predictions may be used for screening, comparison, optimization, or decision support (Bhosekar & Ierapetritou, 2018; de Araujo et al., 2013; Durkin et al., 2024;

Mao et al., 2024).

The scientific machine learning literature offers several ways to connect learned predictions with physical structure. Physics informed training can place governing relations into the loss function, while hard enforcement approaches can use analytic constraints, specialized architectures, conservative formulations, or projection onto physically consistent sets (Beucler et al., 2021; Hansen et al., 2024; Raissi et al., 2019; P. M. Sturm & Wexler, 2020, 2022; Valente et al., 2025). For activated sludge surrogates, the relevant requirement is joint enforcement. A conservation-only correction can still leave negative concentrations, whereas a positivity-only transformation can disturb the sample-specific balance relation. The post prediction projection of Kircher and Votsmeier (2025) is therefore especially relevant because it addresses conservation and nonnegativity together after the surrogate has already produced its component prediction.

The remaining question is how that enforcement behaves in an activated sludge surrogate setting. A correction that guarantees admissibility is valuable only if its empirical consequences are visible: it may improve error when raw residuals point away from the feasible balance relation, but it may also redistribute error among components or composites. The motivating problem is therefore no longer whether machine learning can approximate the simulator at all, but whether outputs from accurate raw surrogates can be converted into physically admissible component states while preserving transparent evidence about predictive error, data scarcity, computational cost, and extrapolation. Paired raw and projected diagnostics are needed to determine whether post prediction enforcement strengthens activated sludge surrogate predictions without treating physical admissibility as a substitute for ordinary validation.

2.2 Method

The analysis evaluated predictive error and physical admissibility through paired raw and projected surrogate outputs. A steady-state activated sludge benchmark supplied complete influent and effluent component vectors. The stoichiometric structure of the mechanistic model was then used to define a sample-specific conservation target and a nonnegative feasible set. Each statistical surrogate first produced an uncorrected 20-component prediction, and the same post prediction projection was then applied to that prediction. This pairing made the effect of projection visible because the raw and projected outputs used the same training rows, validation rows, preprocessing rules, surrogate settings, and mechanistic reference targets.

2.2.1 Simulated Process and Prediction Target

The benchmark data were generated from a standalone, steady-state continuous stirred tank reactor represented by the Activated Sludge Model No. 2d with two-step nitrification (ASM2d-TSN). The process model retained 20 activated sludge components and 28 biological and chemical processes associated with hydrolysis, heterotrophic and phosphate-accumulating-organism metabolism, two-step nitrification and denitrification, lysis, and precipitation-redissolution. The activated sludge model basis followed Henze et al. (2015), while the two-step nitrification formulation, stoichiometry, and rate coefficients followed Guerrero et al. (2011).

Each simulated case used 22 continuous inputs: hydraulic retention time (HRT), aeration, and the 20 influent component concentrations. The effluent target was the corresponding 20-component steady-state vector. The dissolved components were dissolved oxygen S_O , fermentable substrate S_F , acetate S_A , ammonium nitrogen S_{NH4} , nitrite nitrogen S_{NO2} , nitrate nitrogen S_{NO3} , dinitrogen S_{N2} , soluble phosphate S_{PO4} , soluble inert organic matter S_I , and alkalinity S_{ALK} . The particulate components were particulate inert organic matter X_I , slowly biodegradable substrate X_S , heterotrophic biomass X_H , phosphate-accumulating-organism biomass X_{PAO} , stored polyphosphate X_{PP} , stored polyhydroxyalkanoates X_{PHA} , ammonia-oxidizing biomass X_{AOB} , nitrite-oxidizing biomass X_{NOB} , precipitated metal phosphate X_{MeP} , and metal hydroxide X_{MeOH} .

Chemical oxygen demand (COD), total nitrogen (TN), total phosphorus (TP), and total suspended solids (TSS) were computed only after component prediction. For any component vector $c \in \mathbb{R}^F$, with $F = 20$, the reported composite vector was

$$y = (\text{COD}, \text{TN}, \text{TP}, \text{TSS})^\top = I_{\text{comp}} c, \quad (2.1)$$

where $I_{\text{comp}} \in \mathbb{R}^{4 \times F}$ is the fixed component-to-composite composition matrix. The same map was applied to raw and projected component predictions, so the surrogates were not trained directly on the four composites.

The in-distribution design used a randomized, scrambled strength-one Latin hypercube with seed 42 (McKay et al., 1979). The sampled variables and solver rules are summarized in Table 2.1. Each sampled input was solved as a steady-state mechanistic case using nonlinear least squares under componentwise state bounds from 10^{-8} to 1500 in the native unit of each component. Solver cost-change, step-size, and gradient tolerances were 10^{-9} , and each nonlinear solve was limited to 1200 function evaluations. Two physically informed steady-state initializations were attempted first; if both failed, a 20-day stiff backward differentiation formula dynamic relaxation supplied a fallback initializer. Accepted states were required to have a maximum absolute steady-state balance residual no greater than 10^{-9} . The in-distribution dataset retained 10,000 accepted steady states.

Table 2.1: Simulation and acceptance protocol for the surrogate-projection benchmark.

Protocol item	Rule used
Process model	Standalone steady-state continuous stirred tank reactor represented by ASM2d-TSN
Inputs	HRT, aeration, and 20 influent component concentrations
Targets	20 effluent component concentrations in the ASM2d-TSN component basis
Reported composites	COD, TN, TP, and TSS computed from component predictions using I_{comp}
Sampling design	Seed-42 randomized Latin hypercube over the in-distribution operating and influent domain
Accepted in-distribution states	10,000 converged steady-state cases
Steady-state solver	Nonlinear least squares under componentwise box constraints
State bounds	10^{-8} to 1500 in each component's native unit
Solver tolerances	Cost-change, step-size, and gradient tolerances of 10^{-9}
Function evaluations	Maximum of 1200 per nonlinear solve
Fallback initialization	20-day stiff backward differentiation formula dynamic relaxation
Acceptance rule	Maximum absolute steady-state balance residual no greater than 10^{-9}

2.2.2 Conservation Invariants and Feasible Component Space

Let $c_{in,i}, c_{out,i} \in \mathbb{R}^F$ denote the influent and accepted mechanistic effluent component vectors for sample i , and let $\nu \in \mathbb{R}^{P \times F}$ be the ASM2d-TSN Petersen matrix with $P = 28$ processes and $F = 20$ components. The numerical rank of ν was 15, giving a five-dimensional right null space. If the columns of $V_0 \in \mathbb{R}^{F \times K}$, with $K = 5$, form an orthonormal basis for this right null space, then the invariant operator was defined as $A = V_0^\top$. Thus,

$$A\nu^\top = V_0^\top \nu^\top = (\nu V_0)^\top = \mathbf{0}_{K \times P}, \quad AA^\top = V_0^\top V_0 = I_K. \quad (2.2)$$

Each row of A is therefore a linearly independent invariant of the biological and chemical reaction network.

For the adopted steady-state system boundary, the component balance for sample i can be written as

$$D_i(c_{out,i} - c_{in,i}) = \nu^\top \rho_i + \omega_i e_{S_O}, \quad (2.3)$$

where $D_i > 0$ is the hydraulic dilution rate, $\rho_i \in \mathbb{R}^P$ is the vector of biological and chemical process rates evaluated at $c_{out,i}$, and $\omega_i e_{S_O}$ is the external dissolved-oxygen transfer term. Premultiplication by A eliminates the reaction contribution because $A\nu^\top = \mathbf{0}$. The dissolved-oxygen transfer vector is also in the range of $\nu^\top$, so $Ae_{S_O} = \mathbf{0}$ to numerical precision. The balance therefore reduces to the sample-specific invariant relation

$$Ac_{out,i} = Ac_{in,i}. \quad (2.4)$$

The conservation target and feasible effluent set for sample i were defined as

$$b_i := Ac_{in,i}, \quad \mathcal{F}_i = \{c \in \mathbb{R}^F : Ac = b_i, c \geq \mathbf{0}\}. \quad (2.5)$$

The equality in Equation (2.5) defines the five stoichiometric conservation invariants for the stated steady-state boundary, while $c \geq \mathbf{0}$ imposes componentwise nonnegativity. The sampled influent vectors were nonnegative, so each $c_{in,i}$ provided a feasible anchor for projection. In this analysis, mass conservation means satisfaction of these five stoichiometric invariants; it does not claim every possible elemental, kinetic, or plant-wide balance.

2.2.3 Post Prediction Projection

The positivity-preserving conservative projection of Kircher and Votsmeier (2025) was applied after surrogate prediction and did not alter the fitted surrogate. With $u_i \in \mathbb{R}^2$ containing HRT and aeration, the complete input vector was $x_i = [u_i^\top, c_{in,i}^\top]^\top \in \mathbb{R}^{22}$. A trained surrogate $f_m : \mathbb{R}^{22} \rightarrow \mathbb{R}^F$ first produced the raw component prediction

$$c_{raw,i}^{(m)} = f_m(x_i). \quad (2.6)$$

The projection then formed a positive provisional target and a relative scaling matrix,

$$c_{+,i}^{(m)} = \max\left(c_{raw,i}^{(m)}, \epsilon_c\right), \quad s_i^{(m)} = \mathbf{1} \oslash c_{+,i}^{(m)}, \quad W_{s,i}^{(m)} = \text{Diag}\left(s_i^{(m)}\right), \quad (2.7)$$

where the maximum and division are componentwise and each entry of ϵ_c is 10^{-8} in the native unit of its component. The lower-bounded vector $c_{+,i}^{(m)}$ supplied the projection target and its weights; it was not reported as a clipped prediction.

Let $Z \in \mathbb{R}^{F \times (F-K)}$ have orthonormal columns spanning $\text{null}(A)$. Because $Ac_{in,i} = b_i$, every vector satisfying the conservation equality can be represented as

$$\{c : Ac = b_i\} = \{c_{in,i} + Zq : q \in \mathbb{R}^{F-K}\}. \quad (2.8)$$

The projection selected the conservative change Zq closest to the provisional raw-prediction change under the relative scaling in Equation (2.7):

$$q_i^{*,(m)} = \underset{q \in \mathbb{R}^{F-K}}{\text{argmin}} \left\| W_{s,i}^{(m)} \left[Zq - \left(c_{+,i}^{(m)} - c_{in,i} \right) \right] \right\|_2. \quad (2.9)$$

The resulting conservative candidate was

$$c_{cand,i}^{(m)} = c_{in,i} + Zq_i^{*,(m)}. \quad (2.10)$$

This candidate satisfies $Ac_{cand,i}^{(m)} = b_i$. The weighted least-squares system was solved by an orthogonal-triangular factorization, with singular-value decomposition used as a numerical fallback and no normal-equation inverse formed.

The candidate was then shortened toward the feasible influent anchor only when needed to ensure nonnegativity. Let $\delta_i^{(m)} = c_{cand,i}^{(m)} - c_{in,i}$. The backtracking factor and final projected prediction were

$$\alpha_i^{(m)} = \begin{cases} 1, & \min_j c_{cand,ij}^{(m)} \geq 0, \\ (1 - \eta) \min_{j: \delta_{ij}^{(m)} < 0} \frac{c_{in,ij}}{-\delta_{ij}^{(m)}}, & \text{otherwise,} \end{cases} \quad \tilde{c}_i^{(m)} = c_{in,i} + \alpha_i^{(m)} \delta_i^{(m)}, \quad (2.11)$$

with $\eta = 10^{-12}$. The construction preserves conservation because $A\tilde{c}_i^{(m)} = Ac_{in,i} + \alpha_i^{(m)}AZq_i^{*,(m)} = b_i$, and it enforces componentwise nonnegativity by restricting the movement along any direction that would cross the nonnegative boundary. A projection was accepted when $\|A\tilde{c}_i^{(m)} - b_i\|_2 \leq 10^{-10}$ and every component satisfied $\tilde{c}_{ij}^{(m)} \geq -10^{-10}$ in its native unit. Backtracking activation and projection latency were retained as diagnostics.

2.2.4 Surrogates, Preprocessing, and Hyperparameter Selection

Thirteen statistical surrogates were evaluated. The model set matched the raw-prediction comparison and included XGBoost, LightGBM, CatBoost, AdaBoost, Random Forest, Extra Trees, support vector regression (SVR), k -nearest neighbors (k-NN), partial least squares (PLS), Multi-task Elastic Net, Multi-task Lasso, multilayer perceptron (MLP), and TabNet. The set represented boosting, randomized tree ensembles, kernel and neighbor methods, latent-variable regression, sparse multi-task regression, and neural-network surrogates. This selection allowed projection effects to be compared across flexible nonlinear learners, simpler baselines, and more inspectable joint-response models while holding the data and scoring protocol fixed (Arik & Pfister, 2021; Borisov et al., 2024; Breiman, 2001; Chen & Guestrin, 2016; Wold et al., 2001).

Table 2.2: Surrogate groups, output handling, and preprocessing before projection.

Surrogate group	Models	Output handling	Preprocessing before projection and scoring
Boosting	XGBoost, LightGBM, CatBoost, AdaBoost	One regressor per effluent component	XGBoost, LightGBM, and CatBoost used physical input and target units; AdaBoost used target z-scores only
Randomized tree ensembles	Random Forest, Extra Trees	Native multi-output regression	Random Forest used physical input and target units; Extra Trees used target z-scores only
Kernel and neighbor methods	SVR, k-NN	SVR used one regressor per component; k-NN returned all components jointly	Input and target z-scores fitted within the relevant training partition
Latent-variable and sparse multi-task models	PLS, Multi-task Elastic Net, Multi-task Lasso	Joint 20-component regression	Input and target z-scores fitted within the relevant training partition
Neural-network surrogates	MLP, TabNet	Joint 20-component regression	Input and target z-scores fitted within the relevant training partition

All transformations were fitted only on the relevant inner-training, outer-training, or full in-distribution training partition. The fitted transformations were then applied

unchanged to the matching validation or beyond-domain inputs. When a surrogate was trained on target z-scores, its predictions were inverse-transformed to physical component units before projection, conservation evaluation, nonnegativity evaluation, and composite calculation. No missing-value imputation, response-dependent outlier removal, or target filtering was used.

Hyperparameters were selected without access to the scored outer fold. At the largest in-distribution scale, each outer training partition used a four-fold inner search to minimize component-standardized mean squared error. The selected setting for that outer fold was then held fixed when the same surrogate was refitted on the smaller nested training subsets. This made the learning curves a controlled sample-size analysis rather than a sequence of separately optimized small-data studies. Hyperparameter optimization used seeded Optuna studies with a tree-structured Parzen estimator, no pruning, and 100 trials per study (Akiba et al., 2019). A separate full in-distribution search selected the setting used for the final refit before beyond-domain evaluation.

2.2.5 Cross-Validation and Paired Projection Workflow

In-distribution evaluation used nested training-set sizes

$$N \in \{500, 1450, 2400, 3350, 4300, 5250, 6200, 7150, 8100, 9050, 10000\}. \quad (2.12)$$

A seed-42 permutation established both the nested row order and persistent fold assignments before any subset was formed. At every sample size, five-fold cross-validation used the same validation folds for every surrogate. Each fold trained on four fifths of the selected subset and scored the remaining fifth. The common folds made raw-versus-projected comparisons paired by sample, model, and training context.

Table 2.3: Validation, leakage-control, and paired-projection protocol.

Protocol item	Rule used	Data involved	Role in the comparison
Scarcity grid	Eleven nested totals from 500 to 10,000	Seed-42 row permutation	Adds samples without replacing earlier subsets
Outer evaluation	Five folds at every size	0.8N training and 0.2N validation rows	Uses the same folds for every surrogate and prediction type
Inner selection	Four folds within each outer training partition	Largest-scale outer training rows only	Keeps scored outer folds unseen during tuning
Preprocessing	Surrogate-specific scaling	Training partitions only	Prevents validation-response leakage
Inverse transform	Applied before projection	Raw component predictions	Restores physical units for conservation and nonnegativity checks
Paired scoring	Raw and projected versions of one prediction	Identical validation samples	Isolates the effect of projection
Fold summaries	Mean and sample SD across five folds	Fold-level metrics	Describes fold-to-fold variation without hypothesis testing

For each validation row, the fitted surrogate first returned $c_{raw,i}^{(m)}$. The projection then returned $\tilde{c}_i^{(m)}$ using the same $c_{in,i}$, A , Z , and projection tolerances. Raw and projected predictions were scored as paired outputs. The underlying surrogate, pre-processing parameters, training rows, and validation rows were otherwise identical.

2.2.6 Accuracy, Constraint, and Timing Metrics

Component-space accuracy was primary because the 20 effluent components were the supervised targets and because conservation and nonnegativity are defined before aggregation. Let $g \in \{1, \dots, 5\}$ index the outer fold at dataset size N , and let $\mathcal{T}_{N,g}$ be its training partition. For prediction type $t \in \{\text{raw}, \text{proj}\}$, define $c_i^{(m,\text{raw})} = c_{raw,i}^{(m)}$ and $c_i^{(m,\text{proj})} = \tilde{c}_i^{(m)}$. The standardized component residual for validation sample i and component j was

$$\xi_{ij}^{(m,t,N,g)} = \frac{c_{ij}^{(m,t)} - c_{out,ij}}{\sigma_j^{(\mathcal{T}_{N,g})}}, \quad (2.13)$$

where $\sigma_j^{(\mathcal{T}_{N,g})}$ is the population standard deviation of component j in the applicable training partition. For the $n = N/5$ observations in the scored fold, the normalized mean squared error (nMSE), normalized root mean squared error (nRMSE), and normalized mean absolute error (nMAE) were

$$\begin{aligned} \text{nMSE}_{m,t,N,g} &= \frac{1}{Fn} \sum_{i=1}^n \sum_{j=1}^F \left(\xi_{ij}^{(m,t,N,g)} \right)^2, \\ \text{nRMSE}_{m,t,N,g} &= \sqrt{\text{nMSE}_{m,t,N,g}}, \\ \text{nMAE}_{m,t,N,g} &= \frac{1}{Fn} \sum_{i=1}^n \sum_{j=1}^F \left| \xi_{ij}^{(m,t,N,g)} \right|. \end{aligned} \quad (2.14)$$

Component R^2 values and their equal-weight macro-average were computed from residual and total sum-of-squares terms:

$$\begin{aligned} \text{SSE}_{j,m,t,N,g} &= \sum_{i=1}^n \left(c_{ij}^{(m,t)} - c_{out,ij} \right)^2, \\ \text{SST}_{j,N,g} &= \sum_{i=1}^n \left(c_{out,ij} - \bar{c}_{out,j,N,g} \right)^2, \\ R_{j,m,t,N,g}^2 &= 1 - \frac{\text{SSE}_{j,m,t,N,g}}{\text{SST}_{j,N,g}}, \\ \bar{R}_{m,t,N,g}^2 &= \frac{1}{F} \sum_{j=1}^F R_{j,m,t,N,g}^2. \end{aligned} \quad (2.15)$$

Component-specific errors in physical units and composite errors for COD, TN, TP, and TSS were retained as secondary diagnostics. They were not mixed into the primary 20-component standardized score.

Mass-conservation and nonnegativity compliance were assessed before component aggregation. With the unrounded orthonormal basis A , the conservation residual and standardized negative magnitude were

$$v_{mc,i}(c) = \|Ac - b_i\|_2, \quad v_{nn,i}^{\text{std}}(c; \mathcal{T}_{N,g}) = \|\min(c \oslash \boldsymbol{\sigma}^{(\mathcal{T}_{N,g})}, \mathbf{0})\|_1. \quad (2.16)$$

The minimum is componentwise. Conservation and nonnegativity violation frequencies were

$$\begin{aligned} \pi_{mc}^{(m,t,N,g)} &= \frac{1}{n} \sum_{i=1}^n \mathbb{1} \left[v_{mc,i} \left(c_i^{(m,t)} \right) > 10^{-10} \right], \\ \pi_{nn}^{(m,t,N,g)} &= \frac{1}{n} \sum_{i=1}^n \mathbb{1} \left[\exists j : c_{ij}^{(m,t)} < -10^{-10} \right]. \end{aligned} \quad (2.17)$$

Mean, median, and maximum residual magnitudes were retained alongside these frequencies because rare large violations can be hidden by average residuals.

The paired effect of projection on the primary accuracy score and the standardized component displacement were

$$\begin{aligned} \Delta \text{nRMSE}_{m,N,g} &= \text{nRMSE}_{m,\text{proj},N,g} - \text{nRMSE}_{m,\text{raw},N,g}, \\ d_{c,i}^{\text{std},(m)}(\mathcal{T}_{N,g}) &= \left\| \left(\tilde{c}_i^{(m)} - c_{\text{raw},i}^{(m)} \right) \oslash \boldsymbol{\sigma}^{(\mathcal{T}_{N,g})} \right\|_2. \end{aligned} \quad (2.18)$$

Negative ΔnRMSE denotes an accuracy benefit from projection, while positive values denote an accuracy cost. Fold summaries used the five paired fold differences rather than subtracting separately averaged raw and projected scores.

Wall-clock timing was partitioned into preprocessing and surrogate fitting, raw batch inference, projection alone, and end-to-end projected inference. Hyperparameter-search duration was reported separately from fitting and inference. In-distribution timing used a deterministic 512-row validation batch, five warm-up calls, and 30 timed repetitions. The median 512-row latency within a fold was divided by 512 and reported in milliseconds per sample. Projection timing included scaling, the weighted solve, conservation and nonnegativity checks, and positivity backtracking.

2.2.7 Beyond-Domain Evaluation

After cross-validation, one final setting per surrogate was selected using only the 10,000 in-distribution states. Each surrogate and its required preprocessing were then refitted on the complete in-distribution pool before any beyond-domain response was used for scoring. The beyond-domain validation sets were generated with new mechanistic simulations and the same solver, initialization sequence, and acceptance rule used for the in-distribution data. No beyond-domain response was used for refitting, scaling, calibration, or hyperparameter selection.

Four input features were perturbed beyond their in-distribution upper bounds: HRT, aeration, influent ammonium nitrogen, and influent slowly biodegradable substrate. For a perturbed feature ℓ with in-distribution bounds $[L_\ell, U_\ell]$, the normalized

upper-bound exceedance was $(x_{i\ell} - U_\ell)/(U_\ell - L_\ell)$. Mild excursions placed this quantity between 0.10 and 0.25, while severe excursions placed it between 0.35 and 0.60. Candidate generation was dominated by one-feature excursions, with a smaller share of two-feature excursions, and all unperturbed inputs remained inside their in-distribution ranges. Each retained beyond-domain point was therefore outside the in-distribution hyperrectangle.

Table 2.4: Beyond-domain input intervals used for the full-data surrogate refits.

Input	In-distribution interval	Mild interval	Severe interval	Unit
HRT	[6, 36]	[39, 43.5]	[46.5, 54]	h
u_{air}	[0.5, 2.5]	[2.7, 3.0]	[3.2, 3.7]	—
S_{NH4}	[12, 55]	[59.3, 65.8]	[70.1, 80.8]	g N/m ³
X_S	[60, 280]	[302, 335]	[357, 412]	g COD/m ³

Each beyond-domain regime retained 600 accepted steady states. For regime $\zeta \in \{\text{mild}, \text{severe}\}$, standardized errors used the population component standard deviations estimated from all 10,000 in-distribution targets:

$$\xi_{ij}^{(m,t,\zeta)} = \frac{c_{ij}^{(m,t)} - c_{out,ij}}{\sigma_j^{(\mathcal{T}_{in})}}. \quad (2.19)$$

The nMSE, nRMSE, nMAE, macro-average R^2 , violation frequencies, and projection displacements were then computed with the in-distribution reference set replacing the fold-specific training reference. Beyond-domain results were descriptive over 600 cases from one full-data refit per surrogate rather than five-fold cross-validation estimates. Predictive extrapolation and physical admissibility were interpreted separately.

2.2.8 Reproducibility Controls

All pseudorandom streams were derived from root seed 42. The same row order, fold assignments, validation samples, and beyond-domain regimes were used across surrogate families. Seeded fitting was used for XGBoost, LightGBM, CatBoost, Adaboost, Random Forest, Extra Trees, MLP, and TabNet; SVR, k-NN, PLS, and the multi-task linear models were deterministic for fixed data and selected parameters. Double precision was used for mechanistic states, invariant construction, preprocessing, returned predictions, projection, and metrics, except for the internal tensors of TabNet. These controls were intended to make differences between raw and projected outputs attributable to the post prediction projection rather than to changing folds, preprocessing, or random sampling.

2.3 Results

The results are organized around the paired question that motivated the analysis. First, the mechanistic reference states were checked to verify that the benchmark itself occupied the intended physical domain. The surrogate predictions were then evaluated in two forms: raw outputs returned directly by the trained learners and projected outputs obtained after enforcing the declared conservation and nonnegativity conditions.

2.3.1 Benchmark Verification

The completed benchmark contained 10,000 in-distribution steady states and 600 accepted states in each beyond-domain regime. Every mechanistic candidate in these three sets met the acceptance rule (Table 2.5). The accepted effluent states were componentwise positive, and their invariant residuals were small relative to the steady-state acceptance tolerance of 10^{-9} . The largest observed invariant residual was 4.90×10^{-11} in-distribution, 5.67×10^{-11} in the mild beyond-domain regime, and 2.91×10^{-10} in the severe beyond-domain regime.

These checks show that violations observed in raw surrogate outputs were not inherited from infeasible mechanistic targets. The reference states already satisfied the intended component domain before any surrogate was trained. Projection was therefore tested as a post prediction correction of learned outputs, not as a repair of the mechanistic data.

Table 2.5: Mechanistic dataset generation and invariant verification. The invariant residual is $\|Ac_{out} - Ac_{in}\|_2$.

Dataset	Attempted	Retained	Acceptance %	Maximum invariant residual	Minimum component
In-distribution	10,000	10,000	100.0	4.90×10^{-11}	8.75×10^{-4}
Mild beyond-domain	600	600	100.0	5.67×10^{-11}	1.62×10^{-3}
Severe beyond-domain	600	600	100.0	2.91×10^{-10}	1.01×10^{-3}

2.3.2 In-Distribution Accuracy After Projection

At the largest in-distribution sample size, the multilayer perceptron (MLP) retained the lowest raw and projected normalized root mean squared error (nRMSE), followed by CatBoost, LightGBM, XGBoost, and TabNet (Table 2.6). Projection did not change this leading pattern. The best raw surrogates remained the best projected surrogates, which means that post prediction enforcement did not replace the need for a strong learned approximation of the activated sludge response.

Projection reduced aggregate nRMSE for five surrogates and increased it for eight. The largest aggregate benefits occurred for Multi-task Elastic Net and Multi-task Lasso, each with a mean nRMSE change of -0.020 . The strongest aggregate cost occurred for

Random Forest, whose mean nRMSE increased by +0.095. The leading boosted-tree models changed only slightly at full size. CatBoost improved by 0.001, while XGBoost and LightGBM each worsened by about 0.001. The MLP improved by 0.002, indicating that the most accurate raw surrogate also remained accurate after the correction.

Table 2.6: In-distribution performance at $N = 10,000$. Values are five-fold mean $\pm$ sample SD; Δ nRMSE is projected minus raw.

Surrogate	Raw nRMSE	Projected nRMSE	Δ nRMSE	Raw macro- R^2	Projected macro- R^2
XGBoost	0.254 ± 0.006	0.255 ± 0.006	+0.001	0.935 ± 0.003	0.935 ± 0.003
LightGBM	0.249 ± 0.008	0.250 ± 0.007	+0.001	0.938 ± 0.004	0.937 ± 0.004
CatBoost	0.230 ± 0.006	0.229 ± 0.006	-0.001	0.947 ± 0.002	0.947 ± 0.002
AdaBoost	0.525 ± 0.004	0.581 ± 0.028	+0.056	0.724 ± 0.004	0.660 ± 0.035
Random Forest	0.718 ± 0.005	0.813 ± 0.020	+0.095	0.484 ± 0.004	0.332 ± 0.053
Extra Trees	0.563 ± 0.005	0.608 ± 0.025	+0.044	0.683 ± 0.002	0.627 ± 0.037
SVR	0.364 ± 0.011	0.356 ± 0.011	-0.008	0.868 ± 0.002	0.874 ± 0.002
k -NN	0.607 ± 0.007	0.621 ± 0.021	+0.014	0.632 ± 0.001	0.613 ± 0.026
PLS	0.721 ± 0.006	0.745 ± 0.031	+0.024	0.480 ± 0.002	0.446 ± 0.033
Multi-task Elastic Net	0.536 ± 0.008	0.516 ± 0.008	-0.020	0.713 ± 0.003	0.734 ± 0.003
Multi-task Lasso	0.536 ± 0.008	0.516 ± 0.008	-0.020	0.713 ± 0.003	0.734 ± 0.003
MLP	0.150 ± 0.012	0.149 ± 0.012	-0.002	0.977 ± 0.003	0.978 ± 0.003
TabNet	0.272 ± 0.027	0.275 ± 0.024	+0.003	0.926 ± 0.013	0.923 ± 0.014

The learning curves in Figure 2.1 confirm that all raw surrogates benefited from additional mechanistic training states, but not at the same rate. XGBoost and LightGBM were strongest at the smallest sample size, whereas the MLP became the clear full-size leader. TabNet improved sharply from $N = 500$ to $N = 10,000$, but it required more data before becoming competitive with the leading neural and boosted-tree models. These patterns reinforce the importance of separating model learnability from physical enforcement: projection can correct admissibility, but the raw learner still determines the starting point from which the correction operates.

2.3.3 Mass Conservation and Nonnegativity

Every raw surrogate prediction set violated the declared mass-conservation tolerance at every sample size. This included models whose raw predictions remained nonnegative at $N = 10,000$, such as AdaBoost, Random Forest, Extra Trees, and k -NN (Table 2.7). Their nonnegative raw outputs were still not guaranteed to satisfy the case-specific conservation relation $Ac = b_i$. Conversely, the more flexible unconstrained regressors often violated both requirements: at full size, raw nonnegativity violation rates ranged from 39.0% for LightGBM to 76.8% for support vector regression (SVR) among the surrogates that crossed the zero boundary.

Projection eliminated both violation types in all 750,750 in-distribution projected predictions. The largest observed projected conservation residual was 2.06×10^{-13} , and no projected component was negative under the stated tolerance. This is the

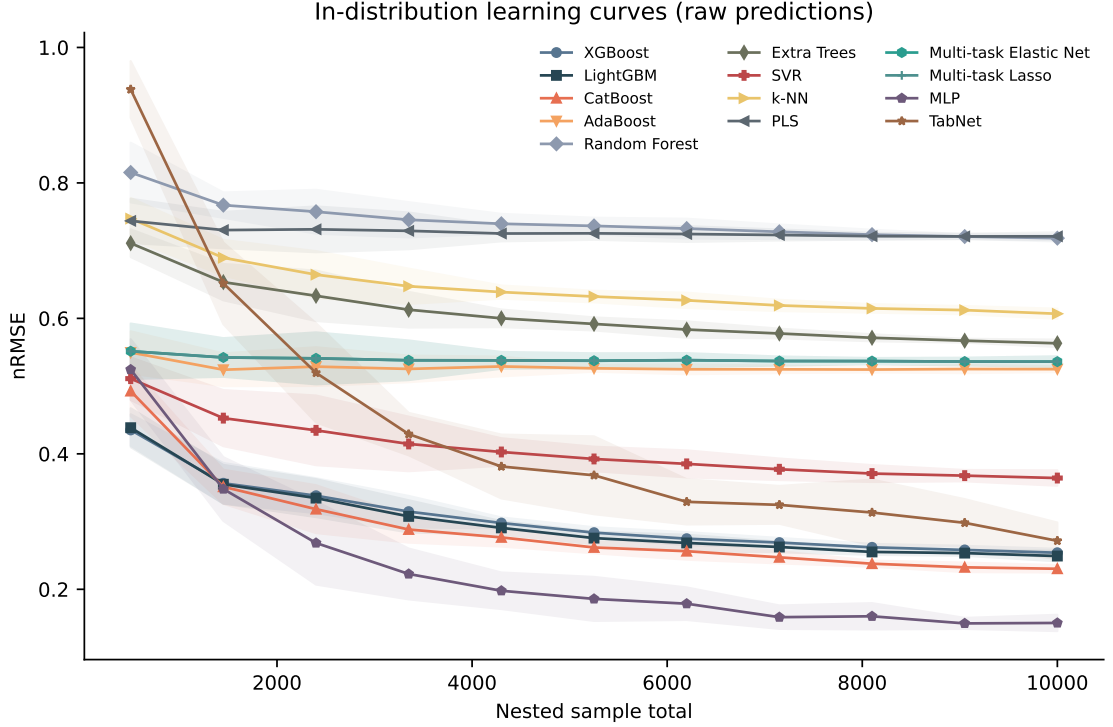


Figure 2.1: In-distribution learning curves for raw predictions. Lines show five-fold mean nRMSE and shaded regions show $\pm$ one sample SD across folds.

central physical-admissibility result: the same post prediction procedure returned every evaluated surrogate output to the declared invariant balance and the nonnegative component domain without retraining the underlying models.

Backtracking was rare. It occurred in 2,489 of the 750,750 in-distribution projections, or 0.332%. Random Forest, partial least squares (PLS), and Extra Trees accounted for most of these activations, with 904, 574, and 566 cases, respectively. The low activation rate indicates that the projection acted primarily as a conservative reconciliation step; positivity backtracking served as a safeguard for the smaller set of predictions whose conservation-preserving candidate would otherwise cross the nonnegative boundary.

The effect of projection on nRMSE varied across models and sample sizes (Figure 2.2). Multi-task Elastic Net and Multi-task Lasso benefited at all eleven sample sizes, suggesting that their joint linear fits left a repeatable residual component outside the feasible conservation subspace. SVR shifted from projection costs at smaller sizes to benefits from $N = 5,250$ onward. The leading boosted trees moved toward near neutrality as the sample size increased, whereas Random Forest retained a clear aggregate cost. This pattern shows why physical enforcement should be evaluated as a paired intervention rather than assumed to improve ordinary accuracy.

Component-level behavior explained why aggregate nRMSE could worsen even when projection usually moved components in a useful direction. At $N = 10,000$,

Table 2.7: Physical-admissibility diagnostics at $N = 10,000$. Raw rates and displacement are five-fold means. Projected mass-conservation and nonnegativity violation rates were zero for every surrogate.

Surrogate	Raw mass violation %	Raw nonnegativity violation %	Mean standardized displacement	Backtracking %
XGBoost	100.0	39.7	0.300	0.00
LightGBM	100.0	39.0	0.301	0.00
CatBoost	100.0	59.1	0.242	0.00
AdaBoost	100.0	0.0	0.901	0.30
Random Forest	100.0	0.0	2.002	1.42
Extra Trees	100.0	0.0	1.528	0.88
SVR	100.0	76.8	0.363	0.01
k -NN	100.0	0.0	1.192	0.25
PLS	100.0	54.3	1.808	0.93
Multi-task Elastic Net	100.0	71.1	0.482	0.00
Multi-task Lasso	100.0	71.1	0.482	0.00
MLP	100.0	58.6	0.189	0.00
TabNet	100.0	60.5	0.545	0.04

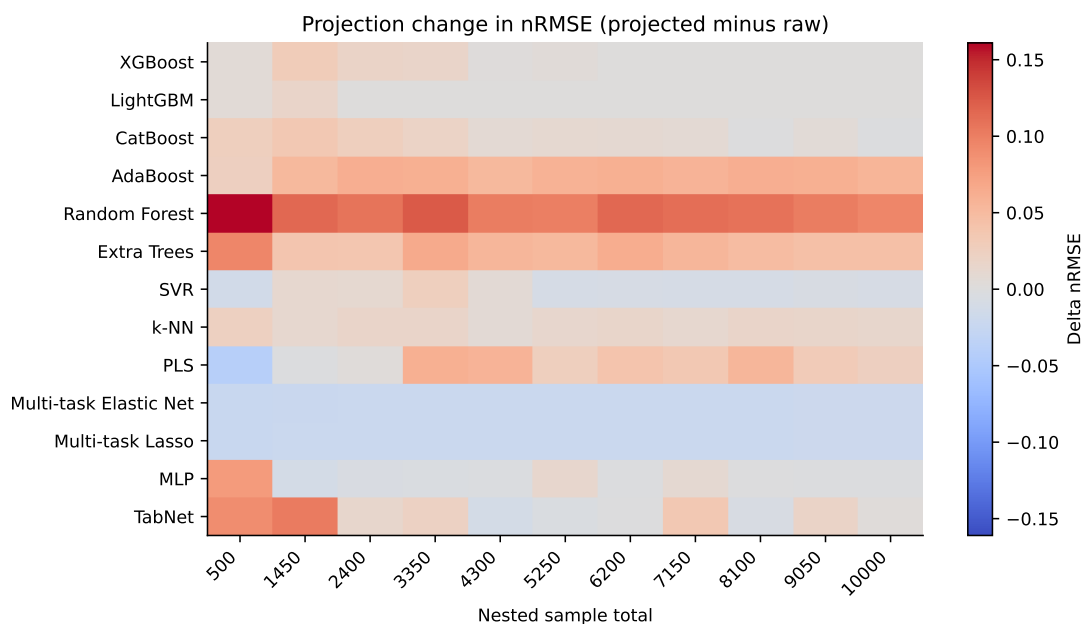


Figure 2.2: Change in in-distribution nRMSE after projection. Each cell is projected minus raw five-fold mean nRMSE; negative values indicate lower error after projection.

projection lowered mean component nRMSE in 188 of 260 surrogate-component comparisons. Twelve of the 13 surrogates improved for a majority of components. Eight components improved for every surrogate: S_{ALK} , S_I , S_{N_2} , S_{NO_2} , S_{PO_4} , X_{MeOH} , X_{MeP} , and X_{NOB} . The principal counterexample was X_H , which worsened for every surrogate. Thus, the aggregate metric was sometimes dominated by a small number of adverse component movements, even though most component-level comparisons improved.

2.3.4 Beyond-Domain Prediction

Beyond-domain evaluation made the prediction problem harder for every raw and projected surrogate (Table 2.8). The MLP remained the lowest-error surrogate in both regimes, with projected nRMSE values of 0.251 under the mild shift and 0.521 under the severe shift. TabNet ranked second in both regimes, followed by the leading boosted-tree models. The shift from mild to severe conditions therefore preserved the broad ordering of the strongest models while increasing the absolute error level.

Projection reduced aggregate nRMSE in 11 of 13 mild-regime comparisons and 9 of 13 severe-regime comparisons. The strongest beyond-domain benefits occurred for the multi-task linear models and PLS, while AdaBoost and Random Forest worsened in both regimes. Extra Trees and k -NN also worsened under severe extrapolation. The result is consistent with the in-distribution pattern: projection can remove infeasible residual directions, but when a raw prediction is far from the mechanistic target, enforcing feasibility can also move selected components farther from their true values.

Table 2.8: Beyond-domain nRMSE for full-data surrogate refits. Each value summarizes 600 separately simulated cases.

Surrogate	Mild raw	Mild projected	Severe raw	Severe projected
XGBoost	0.424	0.420	0.861	0.853
LightGBM	0.419	0.413	0.848	0.842
CatBoost	0.393	0.387	0.836	0.828
AdaBoost	0.692	0.711	1.036	1.139
Random Forest	0.908	0.996	1.242	1.489
Extra Trees	0.717	0.679	1.084	1.222
SVR	0.555	0.538	0.989	0.948
k -NN	0.760	0.732	1.113	1.147
PLS	0.908	0.841	1.267	1.190
Multi-task Elastic Net	0.750	0.688	1.146	1.044
Multi-task Lasso	0.751	0.688	1.146	1.044
MLP	0.255	0.251	0.531	0.521
TabNet	0.336	0.333	0.563	0.547

The component-level beyond-domain pattern again favored projection more often than the aggregate scores alone suggested. Projection lowered component nRMSE in 192 of 260 mild-regime surrogate-component comparisons and 193 of 260 severe-regime comparisons. Every surrogate improved for a majority of components in both regimes. At the same time, adverse movements recurred in a smaller subset of components,

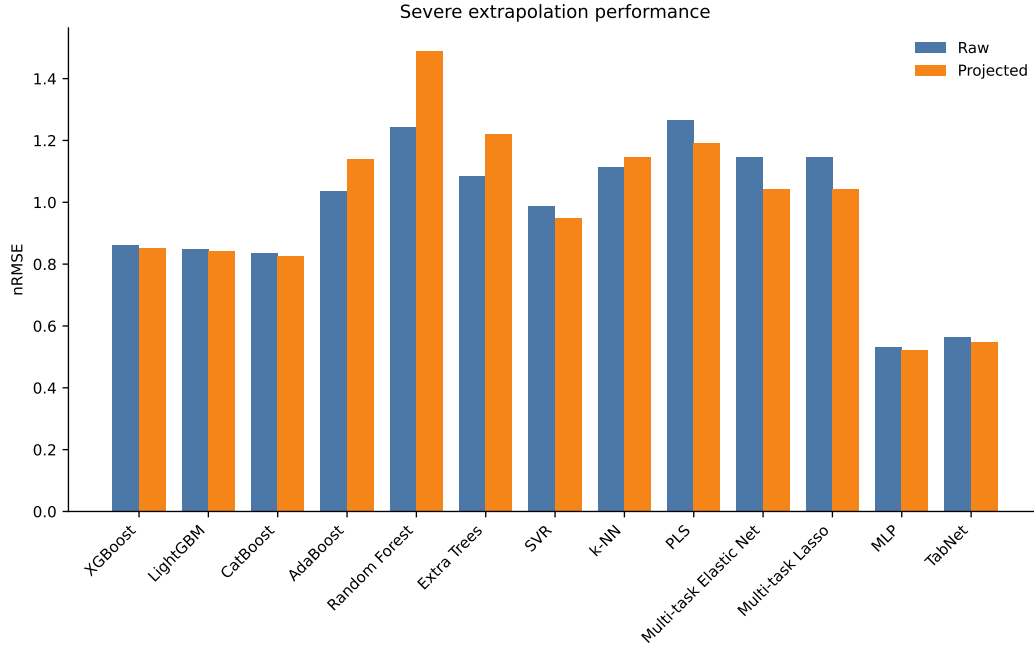


Figure 2.3: Raw and projected nRMSE under the severe beyond-domain regime.

especially X_H and, for several tree and neighbor methods, S_F . These repeated component patterns indicate that projection behavior was governed by the geometry of the invariant operator and the weighting of the correction, not only by the choice of surrogate family.

Every raw beyond-domain prediction set violated mass conservation. Projection reduced both mass-conservation and nonnegativity violation rates to zero across all 15,600 beyond-domain projected predictions. The maximum projected conservation residual was 1.72×10^{-13} , and backtracking was activated for 68 predictions (0.436%). Across in-distribution and beyond-domain evaluation combined, all 766,350 projected predictions satisfied the enforced physical requirements.

2.3.5 Computational Cost

Projection added a small per-sample latency relative to the cost of the complete surrogate workflow (Figure 2.4). At $N = 10,000$, projection-only latency ranged from about 0.068 to 0.144 ms per sample across the surrogate set. End-to-end projected inference remained below 0.315 ms per sample for all models except AdaBoost and SVR. The MLP combined the lowest in-distribution and beyond-domain errors with an end-to-end projected inference latency of 0.107 ms per sample. These timings indicate that enforcing the two physical requirements did not remove the computational advantage of surrogate prediction.

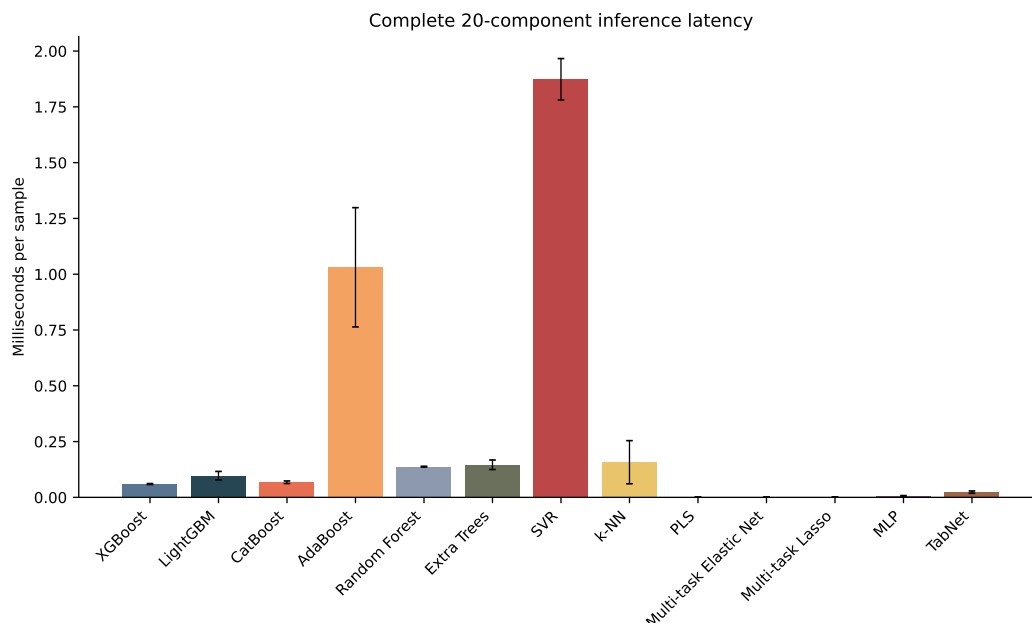


Figure 2.4: Complete 20-component inference latency at $N = 10,000$. Bars show mean raw latency in milliseconds per sample, with fold-to-fold sample SD.

2.4 Synthesis

The results show that physical admissibility is a separate requirement from predictive accuracy. The raw surrogates could learn the activated sludge simulator response with useful accuracy, especially when flexible models were trained on the full in-distribution dataset. Yet every raw prediction set violated the declared mass-conservation condition, and several accurate surrogates also produced negative values in low-concentration components. This decoupling is the main empirical reason for treating ordinary error metrics and physical diagnostics as complementary rather than interchangeable. A low nRMSE indicated closeness to the mechanistic target, but it did not prove that the returned vector could be interpreted as an admissible activated sludge state.

The projection addressed that specific admissibility gap. Across 766,350 projected predictions, it enforced both conservation and nonnegativity with residuals at numerical precision and without changing the underlying surrogate architecture, training loss, or hyperparameter protocol. This matters because component-space predictions are not merely intermediate numerical outputs; they are the basis from which COD, TN, TP, and TSS summaries are derived. If the component state violates a declared balance or contains a negative concentration, then an apparently plausible composite value can hide a physically inconsistent prediction. The paired diagnostics make that problem visible and show that a model-agnostic post prediction step can prevent such outputs from entering downstream interpretation.

The same results also caution against treating projection as a device for improving

accuracy. Projection usually improved component-level scores, but aggregate nRMSE did not improve uniformly. Its effect depended on the position of the raw residual relative to the feasible affine space, the relative weighting in the correction, and the activation of positivity backtracking. The multi-task linear models benefited consistently because projection removed a repeatable inconsistency among components. Random Forest and several averaging models were already nonnegative but farther from the feasible balance relation; correcting them sometimes required larger movements that increased aggregate error. The useful conclusion is not that projection makes every surrogate better. It is that projection makes the physical consequences of each surrogate explicit.

Beyond-domain evaluation reinforces this distinction. Severe shifts increased error for every surrogate, confirming that physical enforcement does not solve extrapolation by itself. Nevertheless, projection often helped under beyond-domain conditions, especially for surrogates whose extrapolation errors contained larger infeasible components. This behavior is consistent with the broader scientific machine learning view that physical structure can constrain learned predictions while still leaving empirical generalization as a separate problem (Hansen et al., 2024; Shen et al., 2023; Valente et al., 2025). For activated sludge surrogates, the practical lesson is that admissibility checks should travel with accuracy checks, particularly when predictions are used outside the center of the sampled operating domain.

Taken together, the evidence supports post prediction projection as a transparent enforcement layer for activated sludge surrogate workflows when complete component states are required. Its strongest contribution is not universal error reduction, but reliable removal of two physically meaningful failure modes: broken stoichiometric invariants and negative concentrations. The limits are equally important. The guarantees apply to the stated steady-state component basis, invariant operator, and nonnegativity condition; they do not prove field validity, kinetic realism, or plant-wide conservation beyond the modeled boundary. Within that scope, the workflow strengthens surrogate credibility by requiring a prediction to be both numerically evaluated and physically interpretable before it is treated as a usable wastewater treatment state.

Chapter 3

A Physically Admissible and Interpretable Machine Learning Surrogate of the Activated Sludge Process

3.1 Rationale

Activated sludge surrogate modeling becomes most useful when the predicted output remains meaningful as a complete process state, not only as a set of accurate numerical estimates. The Activated Sludge Model (ASM) family represents wastewater treatment through coupled component balances for organic matter, nutrients, alkalinity, biomass, and solids, so a predicted effluent vector should remain compatible with the component space in which those balances are defined (Guerrero et al., 2011; Henze et al., 2015). Conventional machine learning surrogates can approximate nonlinear simulator responses with low error, but low error alone does not ensure mass conservation, nonnegativity, or an interpretable relationship between operating conditions, influent composition, and effluent state (Hansen et al., 2024; Kircher & Votsmeier, 2025; Shen et al., 2023). The central motivation is therefore to move from fast prediction toward a surrogate whose returned state is also physically admissible and inspectable.

Two partial solutions leave this need unresolved when they are considered separately. A raw-prediction benchmark can establish whether modern tabular learners recover the activated sludge response from paired inputs and outputs, but the resulting predictions may still be unconstrained component vectors. A post prediction enforcement layer can then return those vectors to the feasible set, but physical admissibility is introduced as a correction after the surrogate has already produced its estimate. That workflow is valuable for diagnosis and repair, yet it does not by itself create a surrogate whose learned structure is organized around the physical roles of influent loads, oper-

ating variables, nonlinear effects, and component coupling. For engineering use, the stronger target is an integrated surrogate design in which predictive accuracy, physical admissibility, and interpretability are treated as connected requirements rather than as separate afterthoughts.

Scientific machine learning provides the broader reason for this integration. Loss functions that encode conservation, analytic constraints, conservative architectures, and projection based corrections show that physical structure can be imposed on learned models, but soft penalties do not necessarily eliminate deployment violations and corrections limited to equality constraints do not necessarily prevent negative states (Beucler et al., 2021; Hansen et al., 2024; Kircher & Votsmeier, 2025; Raissi et al., 2019; P. M. Sturm & Wexler, 2020, 2022; Valente et al., 2025). Activated sludge surrogates require these ideas to be adapted to a component-space prediction problem in which the same effluent vector must support material-balance interpretation, nonnegative concentrations, and derived water quality summaries. The motivating gap is not only the absence of a correction mechanism, but the absence of a transparent surrogate formulation that joins learned nonlinear response with guaranteed final-state admissibility under the declared ASM boundary.

Interpretability makes the gap sharper. In environmental engineering applications, a surrogate may be used for screening operating conditions, comparing scenarios, or supporting optimization, so the model should provide more than a black-box mapping from inputs to outputs when a transparent alternative is feasible. The argument for interpretable models in high-consequence settings is that users should be able to inspect the basis of a prediction rather than rely entirely on post hoc explanation (Rudin, 2019). In the activated sludge setting, this means retaining visible roles for operating conditions, influent component loads, nonlinear curvature, and cross-component redistribution while still enforcing the physical requirements that make the returned vector credible as a process state.

Invariant-Constrained Second-Order Regression (ICSOR) is motivated by this combined requirement. Its role is not simply to replace the most accurate unconstrained learner, nor simply to attach a correction step to an otherwise opaque regression head. The research question is whether a structured, invariant-aware surrogate can provide a practical balance among predictive accuracy, data efficiency, computational cost, interpretability, and exact deployment-time admissibility. Such a balance is important because an activated sludge surrogate that is slightly less flexible but physically dependable and structurally inspectable may be more useful for engineering analysis than a lower-error model whose component predictions require separate repair and remain difficult to interpret.

3.2 Method

The analysis used a steady-state activated sludge benchmark to evaluate whether Invariant-Constrained Second-Order Regression could return predictions that were accurate, physically admissible, and structurally inspectable. All models received the same operating and influent inputs and were trained against the same 20-component effluent state. The methodological distinction was that the comparator regressors learned a direct unconstrained input-output map, whereas ICSOR learned a structured component-space regression head and then returned predictions through a checked deployment rule.

3.2.1 Simulated Process and Prediction Target

The benchmark represented a standalone continuous stirred tank reactor governed by Activated Sludge Model No. 2d with two-step nitrification (ASM2d-TSN). The process description retained 20 activated sludge components and 28 biological and chemical processes, including hydrolysis, heterotrophic growth and denitrification, phosphate-accumulating-organism storage and growth, two-step nitrification, lysis, and phosphorus precipitation-redissolution. The component basis followed the Activated Sludge Model family (Henze et al., 2015), and the two-step nitrification structure followed Guerrero et al. (2011).

Each sample contained two operating variables and the full influent component state. The operating vector $u \in \mathbb{R}^2$ contained hydraulic retention time (HRT) and aeration level. The influent vector $c_{in} \in \mathbb{R}^F$, with $F = 20$, contained dissolved oxygen, soluble carbon, nitrogen, phosphorus, inert, and alkalinity states together with particulate inert, biodegradable substrate, biomass, storage-polymer, and metal-precipitate states. The sampled symbols and ranges are given in Table 3.1. The supervised target was the corresponding effluent component vector $c_{out} \in \mathbb{R}^F$.

Table 3.1: Input ranges used to generate the steady-state ASM2d-TSN benchmark.

Variable	Range	Unit	Variable	Range	Unit
HRT	[6.0, 36.0]	h	S_I	[10.0, 90.0]	g COD/m ³
Aeration	[0.5, 2.5]	—	S_{ALK}	[80.0, 260.0]	mol/m ³
S_O	[0.0, 0.5]	g O ₂ /m ³	X_I	[20.0, 120.0]	g COD/m ³
S_F	[20.0, 180.0]	g COD/m ³	X_S	[60.0, 280.0]	g COD/m ³
S_A	[5.0, 80.0]	g COD/m ³	X_H	[15.0, 100.0]	g COD/m ³
S_{NH4}	[12.0, 55.0]	g N/m ³	X_{PAO}	[5.0, 60.0]	g COD/m ³
S_{NO2}	[0.0, 3.0]	g N/m ³	X_{PP}	[2.0, 20.0]	g P/m ³
S_{NO3}	[0.0, 8.0]	g N/m ³	X_{PHA}	[1.0, 30.0]	g COD/m ³
S_{N2}	[0.0, 2.0]	g N/m ³	X_{AOB}	[0.5, 8.0]	g COD/m ³
S_{PO4}	[2.0, 18.0]	g P/m ³	X_{NOB}	[0.5, 8.0]	g COD/m ³
			X_{MeP}	[0.0, 12.0]	g TSS/m ³
			X_{MeOH}	[0.0, 12.0]	g TSS/m ³

The 22-dimensional input domain in Table 3.1 was sampled by a randomized Latin

hypercube with seed 42 (McKay et al., 1979). Each sampled operating-influent point was solved as a steady-state ASM2d-TSN case by nonlinear least squares under componentwise bounds from 10^{-8} to 1500 in the native unit of each component. The cost-change, step-size, and gradient tolerances were 10^{-9} , and each nonlinear solve was limited to 1200 function evaluations. Physically informed initial guesses were attempted before a dynamic-relaxation fallback was used. A candidate state was retained only when the maximum absolute steady-state balance residual was no greater than 10^{-9} . The final benchmark contained 10,000 accepted steady states.

Chemical oxygen demand (COD), total nitrogen (TN), total phosphorus (TP), and total suspended solids (TSS) were computed after component prediction rather than used as direct training targets. For any component vector c , the reported composite vector was

$$y_{\text{comp}} = I_{\text{comp}}c, \quad (3.1)$$

where $I_{\text{comp}} \in \mathbb{R}^{4 \times F}$ is the fixed component-to-composite composition matrix. Keeping this map outside the learned model ensured that all methods were compared first in the activated sludge component space where conservation and nonnegativity are defined.

3.2.2 Stoichiometric Invariants and Feasible States

Let $\nu \in \mathbb{R}^{28 \times F}$ denote the ASM2d-TSN stoichiometric matrix. A full-row-rank invariant operator $A \in \mathbb{R}^{q \times F}$ was formed from the right null space of ν , so that $A\nu^\top = 0$. The adopted stoichiometric matrix produced $q = 5$ independent invariant rows. For an admissible steady state within the declared reactor boundary, the component change from influent to effluent can be expressed through reaction progress variables ξ , giving

$$c_{\text{out}} - c_{\text{in}} = \nu^\top \xi. \quad (3.2)$$

Premultiplication by A removes the reaction term and gives the conservation relation

$$A(c_{\text{out}} - c_{\text{in}}) = 0, \quad Ac_{\text{out}} = Ac_{\text{in}}. \quad (3.3)$$

This null-space construction follows the same conservation logic used in physically structured surrogate models, where learned predictions are required to remain on a constraint manifold rather than only near the observed target values (P. M. Sturm & Wexler, 2020, 2022).

For sample i , the conservation target and feasible effluent set were

$$b_i = Ac_{\text{in},i}, \quad \mathcal{F}_i = \{c \in \mathbb{R}^F : Ac = b_i, c \geq \mathbf{0}\}. \quad (3.4)$$

The equality condition encodes the five modeled stoichiometric invariants, and the inequality condition requires componentwise nonnegativity. These constraints define physical admissibility for the returned component vector. They do not claim complete kinetic realism, thermodynamic feasibility, or plant-wide conservation outside the adopted system boundary.

3.2.3 Invariant-Constrained Second-Order Regression

ICSOR separates the operating variables from the influent inventory when constructing its regression head. For each sample, the feature vector was

$$\phi(u, c_{in}) = \begin{bmatrix} 1 \\ u \\ c_{in} \\ u \otimes u \\ c_{in} \otimes c_{in} \\ u \otimes c_{in} \end{bmatrix} \in \mathbb{R}^D, \quad D = 1 + M_{op} + F + M_{op}^2 + F^2 + M_{op}F, \quad (3.5)$$

with $M_{op} = 2$. This partition keeps the engineering roles of operating settings and influent loads explicit. Operating variables alter the reactor environment, whereas influent concentrations define the material inventory supplied to that environment (de Araujo et al., 2013; Machado et al., 2014).

The feature vector drives a latent component-space forcing vector rather than the final deployed prediction directly:

$$r(u, c_{in}) = B\phi(u, c_{in}). \quad (3.6)$$

Written by blocks, this driver is

$$r = b + W_u u + W_{in} c_{in} + \Theta_{uu}(u \otimes u) + \Theta_{cc}(c_{in} \otimes c_{in}) + \Theta_{uc}(u \otimes c_{in}). \quad (3.7)$$

The blocks respectively represent baseline forcing, first-order operating effects, first-order influent-load effects, operating curvature, influent-state curvature, and operation-loading interactions. After each update of B , the self-quadratic blocks were symmetrized so that duplicated ordered products such as $x_i x_j$ and $x_j x_i$ had the same fitted contribution. This step preserves fitted values while making the quadratic coefficient blocks easier to inspect.

Cross-component dependence was represented through a learned coupling matrix $\Gamma \in \mathbb{R}^{F \times F}$. With $R = I_F - \Gamma$, the coupled regression relation was

$$Rc = r(u, c_{in}). \quad (3.8)$$

The diagonal of Γ was fixed to zero, its off-diagonal entries were bounded, and candidate updates were accepted only when R remained well conditioned. Thus, the regression head could represent redistribution among activated sludge components without allowing deployment to depend on a nearly singular coupling matrix.

For a training set with feature matrix Φ , influent matrix C_{in} , and effluent target matrix C_{out} , ICSOR estimated B , Γ , and an auxiliary nonnegative fitted-state matrix $\hat{C}$ by minimizing

$$\begin{aligned} (\hat{B}, \hat{\Gamma}, \hat{C}) = \underset{B, \Gamma \in \mathcal{G}, \hat{C} \in \mathbb{R}_+^{N \times F}}{\operatorname{argmin}} \quad & \left\| C_{out} - \hat{C} \right\|_F^2 + \lambda_{inv} \left\| (\hat{C} - C_{in}) A^\top \right\|_F^2 \\ & + \lambda_{sys} \left\| \hat{C} (I_F - \Gamma)^\top - \Phi B^\top \right\|_F^2 + \lambda_B \|B\|_F^2 + \lambda_\Gamma \|\Gamma\|_F^2. \end{aligned} \quad (3.9)$$

The first term fits the effluent components directly. The invariant term is a soft training penalty rather than a hard guarantee. The coupled-system term links the fitted states to the structured regression head, and the ridge penalties stabilize the driver and coupling estimates. Exact conservation and nonnegativity are therefore properties of the deployment rule described below, not of the soft training objective alone.

The objective was minimized through deterministic block-coordinate updates. For fixed Γ and $\widehat{C}$, the B update was a ridge-regularized least-squares problem. For fixed B and $\widehat{C}$, the Γ update was a convex box-constrained quadratic program within the admissible coupling set $\mathcal{G}$. For fixed B and Γ , the $\widehat{C}$ update was a nonnegative quadratic program. Each full cycle updated the three blocks and retained the best objective value reached under the selected stopping rules.

Table 3.2: Selected ICSOR settings used in the benchmark.

Setting	Selected value
Bias term in ϕ	Included
λ_{inv}	8.68×10^{-3}
λ_{sys}	1.27×10^{-3}
λ_B	6.09×10^{-5}
λ_Γ	6.55×10^{-4}
Maximum absolute off-diagonal Γ entry	0.368
Maximum outer iterations	60
Restarts	1
Objective-regression window	12 iterations
Objective-regression slope tolerance	2.85×10^{-7}

3.2.4 Checked Deployment

For a new input pair $(u_*, c_{in,*})$, the fitted driver and coupling matrix first define

$$\phi_* = \phi(u_*, c_{in,*}), \quad d_* = \widehat{B}\phi_*, \quad \widehat{R} = I_F - \widehat{\Gamma}. \quad (3.10)$$

When $\widehat{R}$ is nonsingular, the raw coupled state is

$$c_{raw} = \widehat{R}^{-1}d_*. \quad (3.11)$$

This state is the transparent regression-head prediction, but it is not automatically physically admissible. The deployment rule therefore checks the raw state and, when needed, applies a staged correction.

The first correction stage is the nearest invariant-consistent affine projection:

$$c_{aff} = \underset{c \in \mathbb{R}^F}{\operatorname{argmin}} \|c - c_{raw}\|_2^2 \quad \text{subject to} \quad Ac = Ac_{in,*}. \quad (3.12)$$

Because A has full row rank, the closed-form solution is

$$c_{aff} = c_{raw} - A^\top (AA^\top)^{-1} A(c_{raw} - c_{in,*}). \quad (3.13)$$

This stage restores the invariant relation but does not by itself guarantee nonnegative component concentrations.

The final branch is activated only when the raw or affine state fails the feasibility checks. It solves

$$\begin{aligned} \hat{c} = & \underset{c \in \mathbb{R}^F, \rho \in \mathbb{R}^F}{\operatorname{argmin}} && w_c^\top \rho \\ \text{subject to} & && Ac = Ac_{in,*}, \\ & && c \geq \mathbf{0}, \\ & && -\rho \leq c - c_{aff} \leq \rho, \\ & && \rho \geq \mathbf{0}, \end{aligned} \tag{3.14}$$

where $w_c \in \mathbb{R}_+^F$ is a fixed vector of componentwise correction weights. The program keeps the final state close to the affine reference in a weighted L_1 sense while imposing both conservation and nonnegativity. The feasible set is nonempty because the nonnegative influent state itself satisfies $Ac = Ac_{in,*}$ when $c = c_{in,*}$. This checked deployment logic follows the same practical lesson emphasized by Kircher and Votsmeier (2025): a conservative correction is physically useful only when positivity is preserved at the same time.

Deployment followed a hierarchy. If c_{raw} satisfied $Ac_{raw} = Ac_{in,*}$ and was nonnegative within the numerical tolerance, it was returned directly. If the raw state failed but c_{aff} was nonnegative, the affine state was returned. Otherwise, Equation (3.14) returned the final deployed state $\hat{c}$. Reported composites were then computed as $I_{\text{comp}}\hat{c}$.

3.2.5 Benchmark Models, Tuning, and Validation

ICSOR was compared with nine unconstrained regressors: XGBoost, LightGBM, CatBoost, AdaBoost, Random Forest, support vector regression (SVR), k -nearest neighbors (k-NN), partial least squares (PLS), and a fully connected multilayer perceptron (MLP). These comparators represented boosting, randomized trees, kernel regression, instance-based prediction, latent-variable regression, and neural-network regression (Borisov et al., 2024; Breiman, 2001; Chen & Guestrin, 2016; Wold et al., 2001). The MLP used three hidden layers with widths 128, 64, and 32 and mapped the 22-dimensional input vector to the 20-component effluent state.

Hyperparameters for all retained models were selected by seeded Optuna studies with a tree-structured Parzen estimator sampler, no pruning, and 100 trials per model (Akiba et al., 2019). Selection minimized validation mean squared error in the shared 20-component effluent prediction space. The 8,000-row training pool from the fixed benchmark split was reduced to a 4,000-row tuning subset, and that subset was split into 3,200 tuning-training rows and 800 tuning-validation rows. The selected settings were then fixed for the one-shot benchmark and the repeated dataset-size analysis.

A row-wise 80–20 train-test split with seed 42 supplied the main benchmark, yielding 8,000 training rows and 2,000 test rows. The unconstrained regressors used z-score

transformations fitted on the training split only, separately for the 22-input matrix and 20-output component matrix, and their predictions were inverse transformed before any physical diagnostic or composite metric was computed. ICSOR was kept in physical coordinates so that the invariant operator, deployment rule, and composition map remained aligned with the component basis. This preprocessing rule avoided leakage from the held-out test set (Pinheiro et al., 2025).

Sample efficiency and computational scaling were evaluated with a repeated size study. The retained totals were

$$n \in \{500, 1450, 2400, 3350, 4300, 5250, 6200, 7150, 8100, 9050, 10000\}. \quad (3.15)$$

At each size, ten independent subsamples were drawn without replacement from the 10,000 accepted states using seeds derived from root seed 42. Each subsample was split row-wise into 80% training and 20% held-out test rows. The selected hyperparameters were not retuned during this sweep, so the resulting learning curves isolated sample-size sensitivity rather than repeated tuning variability.

3.2.6 Evaluation Metrics and Interpretability Diagnostics

Predictive accuracy was reported after every model’s 20-component prediction was converted to COD, TN, TP, and TSS through the common composition matrix. For a model m , target composite k , and held-out sample set $\mathcal{I}$, the composite residual was

$$e_{ik}^{(m)} = y_{ik}^{(m)} - y_{ik}^{\text{ref}}. \quad (3.16)$$

The aggregate and per-target summaries included root mean squared error (RMSE), mean absolute error (MAE), mean absolute percentage error (MAPE), mean squared error (MSE), and the coefficient of determination R^2 . The fixed split used aggregate RMSE as the primary ordering metric because it measures error in the same reported-composite space used for engineering summaries.

For the main accuracy comparison, ICSOR was evaluated at its final checked deployed output. The comparator regressors were evaluated after the benchmark’s common equality-only affine alignment and the same external composition map. This alignment enforced the reporting invariants but did not impose componentwise nonnegativity and was not the ICSOR linear program. Physical-admissibility diagnostics were reported separately before composite collapse so that numerical alignment for accuracy would not be mistaken for a learned guarantee of feasibility.

For a component prediction $\tilde{c}_i$, mass-conservation and nonnegativity violations were quantified as

$$v_{mc,i} = \|A(\tilde{c}_i - c_{in,i})\|_2, \quad v_{nn,i} = \|\min(\tilde{c}_i, \mathbf{0})\|_1, \quad (3.17)$$

where the minimum is componentwise. A sample was counted as violating mass conservation or nonnegativity when the corresponding quantity exceeded 10^{-10} . ICSOR

diagnostics distinguished the raw coupled state, the affine projected state, and the final checked deployed state. Comparator diagnostics used their direct raw component predictions.

Learning behavior across sample sizes was summarized with final-size RMSE, normalized area under the RMSE learning curve, dense average rank across retained targets and metrics, and the train-validation RMSE gap at the largest retained size. The normalized area under the learning curve (nAUC) was computed by trapezoidal integration over the sampled dataset sizes:

$$\text{nAUC}_{i,m} = \frac{1}{n_{\max} - n_{\min}} \int_{n_{\min}}^{n_{\max}} \overline{M}_{i,m}(n) dn, \quad (3.18)$$

where $\overline{M}_{i,m}(n)$ is the mean held-out value of metric m for model i at dataset size n . This summary makes the full learning curve visible rather than ranking models only at the largest sample size (Viering & Loog, 2023). Wall-clock training time was recorded for every repeated fit.

Interpretability was evaluated on the globally defined pre-deployment regression head rather than on the piecewise correction returned by checked deployment. For reported composites, the raw-head coefficient map was

$$B_{\text{raw,comp}} = I_{\text{comp}} \hat{R}^{-1} \hat{B}. \quad (3.19)$$

The named blocks of this matrix were then inspected to identify operating effects, influent effects, curvature within each input group, and interactions between operation and loading. The coupling matrix $\hat{\Gamma}$ was inspected separately as the learned redistribution structure among activated sludge components. Coefficient interpretation was therefore limited to structural patterns in the raw head and coupling matrix; it was not treated as causal evidence about the biological process or as a local explanation of the final deployed correction (Rudin, 2019; Shen et al., 2023).

3.3 Results

3.3.1 Fixed-Split Predictive Performance

Table 3.3 summarizes the fixed 80–20 test benchmark after each predicted component state was converted to COD, TN, TP, and TSS through the common composition matrix. The lowest aggregate test error was obtained by the MLP, with RMSE 4.38, MAE 2.55, $R^2 = 0.9915$, and MAPE 0.0123. LightGBM followed with RMSE 5.30. SVR and ICSOR were nearly tied, with RMSE values of 5.97 and 5.98, respectively. XGBoost and CatBoost followed at 6.12 and 6.45. The remaining models formed a higher-error group, with aggregate RMSE values from 21.27 to 29.00.

The fixed-split comparison therefore shows a moderate accuracy cost for the physically checked ICSOR output. Its aggregate RMSE was 36.5% higher than the MLP

Table 3.3: Fixed-split 80–20 test performance in the reported-composite space. ICSOR is evaluated after checked deployment; comparator predictions use the common equality-only affine alignment before composite calculation.

Model	RMSE	MAE	R^2	MAPE
MLP	4.38	2.55	0.9915	0.0123
LightGBM	5.30	3.21	0.9806	0.0203
SVR	5.97	3.49	0.9713	0.0244
ICSOR	5.98	3.70	0.9700	0.0256
XGBoost	6.12	3.81	0.9733	0.0244
CatBoost	6.45	4.07	0.9727	0.0250
AdaBoost	21.27	13.15	0.8643	0.0653
k -NN	23.61	13.87	0.8535	0.0557
PLS	28.24	17.05	0.7808	0.0681
Random Forest	29.00	16.93	0.7934	0.0632

value and 12.8% higher than the LightGBM value. At the same time, ICSOR remained in the leading group rather than in the high-error group. Its RMSE was essentially the same as SVR and lower than XGBoost and CatBoost under the stated reporting protocol.

The per-target decomposition in Table 3.4 indicates that the aggregate ordering was driven mainly by COD, TN, and TSS. The MLP had the lowest RMSE for COD, TN, and TSS, while LightGBM was the strongest non-neural model on those same composites. ICSOR was competitive on COD, with RMSE 7.77, which was 0.18 higher than LightGBM and lower than XGBoost, CatBoost, SVR, and the remaining baselines. Its larger TN and TSS errors prevented that COD performance from producing a better aggregate rank. TP did not distinguish the retained models because all equality-aligned reported outputs had RMSE 0.0361 for that composite.

Table 3.4: Per-target fixed-split test RMSE in the reported-composite space.

Model	COD	TN	TP	TSS
MLP	6.48	1.58	0.0361	5.66
LightGBM	7.59	2.77	0.0361	6.85
SVR	8.87	3.50	0.0361	7.20
ICSOR	7.77	3.55	0.0361	8.36
XGBoost	8.78	3.27	0.0361	7.88
CatBoost	9.13	3.23	0.0361	8.53
AdaBoost	38.30	5.58	0.0361	17.68
k -NN	38.24	4.23	0.0361	27.37
PLS	43.22	5.09	0.0361	36.00
Random Forest	45.87	4.00	0.0361	35.26

3.3.2 Sample Efficiency, Generalization, and Training Cost

The repeated size study changed the interpretation of the fixed-split ranking. At the smallest retained dataset size of 500 rows, ICSOR had the lowest mean held-out

RMSE at 10.79. XGBoost and LightGBM followed at 11.26 and 11.45, while the MLP ranked fifth at 13.26. At the largest retained size, the ordering shifted: the MLP had the lowest mean held-out RMSE at 4.62, followed by LightGBM at 5.27, SVR at 5.84, and ICSOR at 5.94. This crossover is consistent with the broader learning-curve expectation that structurally restricted models can be sample efficient at small N , while more flexible models may overtake them when enough data are available (Shen et al., 2023; Viering & Loog, 2023).

Table 3.5 condenses the repeated size sweep. ICSOR had the lowest RMSE nAUC at 6.33, narrowly ahead of LightGBM at 6.35. This indicates that ICSOR’s average RMSE profile across the retained sample-size range was favorable even though it did not have the lowest final-size RMSE. LightGBM had the best sample-average rank at 2.2, followed by the MLP at 2.5. ICSOR and XGBoost were tied at 4.3, showing that ICSOR’s low-data advantage did not produce a dominant average rank across all retained metrics, targets, and sample sizes.

Table 3.5: Summary of the repeated size benchmark. Final RMSE and Δ RMSE are evaluated at the largest retained size. Lower values are better for all columns.

Model	Final RMSE	RMSE nAUC	Sample avg. rank	Δ RMSE
MLP	4.62	6.75	2.5	0.65
LightGBM	5.27	6.35	2.2	1.99
SVR	5.84	6.90	4.0	1.19
ICSOR	5.94	6.33	4.3	0.22
XGBoost	6.10	7.01	4.3	1.44
CatBoost	6.26	7.60	5.0	0.69
AdaBoost	21.42	20.29	8.2	0.23
k -NN	23.76	25.77	7.0	23.66
PLS	27.96	30.17	9.4	0.38
Random Forest	28.78	30.31	8.2	18.71

The train-validation RMSE gap gives a separate view of generalization. At the largest retained size, ICSOR’s gap was 0.22, lower than the gaps for LightGBM, XGBoost, SVR, and the MLP. PLS also had a small gap at 0.38, but its absolute errors were high. Thus, the small gap for ICSOR reflected relatively close train and held-out performance among the comparatively accurate models, whereas the small gap for PLS was consistent with underfitting.

This generalization behavior came with additional fitting cost. At 500 rows, ICSOR required 21.5 seconds of mean training time, compared with 4.72 seconds for LightGBM, 3.16 seconds for XGBoost, and 4.99 seconds for the MLP. At 10,000 rows, ICSOR required 94.4 seconds. That was slower than the tree-based models, PLS, and k -NN, but faster than AdaBoost, SVR, and the MLP at the same size. The recursive coupled-QP estimator was therefore not the fastest option, but it remained within the range of practical offline surrogate fitting.

3.3.3 Physical Admissibility and Deployment Behavior

Physical-admissibility diagnostics were evaluated before the predicted component state was collapsed to COD, TN, TP, and TSS. Table 3.6 reports the final checked ICSOR output alongside the direct raw-head outputs of the unconstrained comparators. ICSOR was the only retained pipeline with 0.0% mass conservation violations and 0.0% nonnegativity violations. Every unconstrained comparator violated the stoichiometric conservation relation on all test samples. Nonnegativity behavior varied: AdaBoost, Random Forest, and k -NN produced no negative component concentrations on the test set, while the remaining unconstrained models had nonnegativity violation rates from 46.00% to 70.10%.

Table 3.6: Physical-admissibility violation frequencies on the fixed test split. Mass conservation and nonnegativity are assessed before external composite calculation.

Model	Mass conservation violations (%)	Nonnegativity violations (%)
MLP	100.0	58.20
LightGBM	100.0	46.00
SVR	100.0	68.65
ICSOR	0.0	0.00
XGBoost	100.0	64.95
CatBoost	100.0	70.10
AdaBoost	100.0	0.00
k -NN	100.0	0.00
PLS	100.0	48.70
Random Forest	100.0	0.00

The models with zero nonnegativity violations were not physically admissible in the full component-space sense. Their outputs remained in the nonnegative orthant because of their averaging or aggregation geometry, but they still broke the sample-specific conservation relation on every test case. Conversely, several of the most accurate models in Table 3.3, including the MLP and LightGBM, violated both conditions. Accuracy in the reported-composite space therefore did not imply that the underlying component vector could be interpreted as an admissible activated sludge state. This pattern agrees with the scientific machine learning concern that physically meaningful constraints often need explicit enforcement rather than passive expectation from low prediction error (Hansen et al., 2024; Kircher & Votsmeier, 2025; P. O. Sturm & Silva, 2024).

Table 3.7 separates the stages of the ICSOR deployment rule. None of the raw coupled predictions satisfied both conservation and nonnegativity. The raw head was nonnegative for 21.7% of test samples, but raw conservation compliance was 0.0%. The affine stage restored conservation for all test samples and was sufficient for 436 of the 2,000 test cases, or 21.8%. The remaining 1,564 cases required the linear programming branch. The final returned output was feasible for all 2,000 test samples.

The linear programming branch converged in a mean of 23.5 HiGHS iterations, with

Table 3.7: ICSOR deployment-stage behavior on the fixed test split.

Deployment stage	Conservation compliant (%)	Nonnegative (%)	Returned samples
Raw coupled state	0.0	21.7	0
Affine invariant projection	100.0	21.8	436
Linear programming correction	100.0	100.0	1,564
Final checked output	100.0	100.0	2,000

a maximum of 40 iterations. Thus, the hard feasibility guarantee came from checked deployment rather than from the soft invariant penalty in the training objective alone. The affine stage also had a practical role: it resolved the subset of samples whose invariant correction did not create negative components, reserving the general optimizer for cases in which nonnegativity constraints were active.

3.3.4 Interpretable Raw-Head Structure

The interpretability analysis was performed on the globally defined pre-deployment raw head. Since the final affine and linear-programming stages are active-set dependent, the checked deployed output does not have one global coefficient matrix. The raw-head coefficient map $B_{\text{raw,comp}} = I_{\text{comp}} \hat{R}^{-1} \hat{B}$ was therefore used to inspect the reported-composite structure associated with the fitted ICSOR regression head.

Table 3.8 summarizes the retained coefficient counts for the COD raw-head structure. The reported-output blocks used a threshold equal to 10^{-4} of the largest absolute COD raw-head coefficient, while the shared Γ block used 10^{-4} of its own largest off-diagonal magnitude. The threshold was applied only for reporting; it did not affect fitting, model selection, or deployment.

Table 3.8: Retained coefficients in the COD raw-head structure. Symmetric quadratic totals count unique upper-triangular terms.

Coefficient block	Total coefficients	Retained coefficients	Retained (%)
b	1	1	100.0
W_u	2	2	100.0
W_{in}	20	19	95.0
Θ_{uu}	3	3	100.0
Θ_{uc}	40	29	72.5
Θ_{cc}	210	44	21.0
Γ	380	371	97.6
Overall	656	469	71.5

The COD raw-head structure retained all operating terms, nearly all first-order influent terms, and all operating-curvature terms. It retained 29 of 40 operation-loading interaction terms and 44 of 210 unique influent-curvature terms. This pattern suggests that the fitted COD raw head used broad first-order loading information and operating effects, but selected a smaller subset of nonlinear influent-state curvature

terms. The coupling matrix was comparatively dense, with 371 of 380 off-diagonal entries retained under the reporting threshold. The resulting model was therefore not sparse in the simple sense. Its interpretability came from the named block structure and separable coupling representation rather than from a small number of nonzero coefficients.

The largest COD influent-curvature terms involved dissolved oxygen and nitrogen or phosphorus-metal states. The strongest self-curvature was the S_O - S_O term at -5.8097 , followed by the mirrored S_O - S_{N2} pair at -1.7411 . Other prominent terms linked S_O with X_{MeP} , X_{AOB} , S_{PO4} , X_{NOB} , and S_{NO2} . These associations are consistent with the coupled oxygen, nitrogen, and phosphorus behavior represented in activated sludge nutrient-removal models (de Araujo et al., 2013; Henze et al., 2015; Wentzel et al., 1992). They should be interpreted as structural patterns in the fitted raw head, not as causal effects or as derivatives of the final deployed output.

3.4 Synthesis

The results show why an integrated surrogate is a distinct contribution rather than a repetition of raw prediction benchmarking or post prediction repair. The fixed-split accuracy results confirm that the most flexible unconstrained learner, the MLP, gave the lowest reported-composite error. LightGBM also outperformed ICSOR on aggregate RMSE. If the sole objective were minimum error in COD, TN, TP, and TSS, those models would be preferable within the evaluated benchmark. ICSOR instead occupied a different position: it remained close to the leading accuracy group while adding a deployment contract that the comparator regressors did not provide.

The physical diagnostics are the strongest evidence for that distinction. Every unconstrained raw head violated the stoichiometric conservation relation on every test sample, even when its reported-composite error was low or its component predictions were nonnegative. This means that statistical accuracy and component-space admissibility were empirically separate properties. ICSOR closed that gap only after checked deployment: the raw regression head alone was not feasible, and the affine projection alone resolved only part of the test set. The final deployment branch was therefore not an incidental correction but the mechanism that converted a structured regression estimate into a usable activated sludge state.

The repeated size analysis adds a second qualification to the accuracy tradeoff. ICSOR was strongest at the smallest retained dataset size and had the lowest RMSE nAUC across the full sample-size sweep. Its train-validation gap was also small among the comparatively accurate models. These findings suggest that the model's structural restrictions helped when data were limited and reduced train-test separation at the largest size. The same restrictions also limited final-size accuracy relative to the MLP and LightGBM. The most defensible interpretation is therefore a tradeoff among flexi-

bility, sample efficiency, and physical dependability rather than a universal ranking of model families.

The interpretability results further clarify the role of ICSOR. The model did not become interpretable because every coefficient was sparse or causal. Instead, its raw head exposed named blocks for operating effects, influent effects, within-group curvature, operation-loading interaction, and cross-component coupling. This structure is more inspectable than an opaque direct map, but it still requires caution: the coefficients belong to the pre-deployment head, and the final checked output is shaped by affine and linear-programming correction when feasibility constraints are active. Interpretation should therefore focus on recurring structural patterns rather than isolated coefficient magnitudes.

Within the stated steady-state reactor boundary, ICSOR provides a practical balance of accuracy, data efficiency, physical admissibility, and structural inspection. Its evidence is limited to simulated ASM2d-TSN component states within the sampled operating and influent ranges, and its conservation guarantee is only as broad as the stoichiometric invariants used to define the feasible set. It does not establish field validity, kinetic realism, or plant-wide conservation beyond the modeled boundary. The contribution is more specific and more useful: it shows that an activated sludge surrogate can be designed so that the returned component vector is not merely numerically plausible, but also exactly conservative, nonnegative, and connected to an inspectable regression structure.

Conclusion

Summary of Contributions

This research examined how machine learning surrogates of the activated sludge process can be developed and evaluated so that their predictions are not only fast and accurate, but also physically admissible and interpretable. Mechanistic activated sludge models represent biological wastewater treatment through coupled component balances for organic carbon, nitrogen, phosphorus, alkalinity, biomass, and solids, yet repeated simulation of these models is costly when many operating and influent conditions must be explored (Guerrero et al., 2011; Henze et al., 2015). Data driven surrogates promise to relieve that cost, but low prediction error alone does not guarantee that a returned component vector can be read as a valid process state (Hansen et al., 2024; Kircher & Votsmeier, 2025). The work addressed this concern through three connected studies that treated predictive accuracy, mass conservation, nonnegativity, and structural transparency as joint requirements rather than as separate goals.

The first study established a controlled surrogate benchmark for the complete effluent component state of a steady-state Activated Sludge Model No. 2d with two-step nitrification (ASM2d-TSN) simulator. By comparing thirteen tabular learners under shared sampling, folds, and scoring rules, it showed that the activated sludge response is learnable from paired inputs and outputs, but that raw accuracy depends on the model family, the amount of training data, the identity of the effluent component, and the degree of domain shift. Boosted trees were the most data efficient at the smallest sample size, whereas a multilayer perceptron (MLP) used the largest training set most effectively and achieved the lowest error and the highest component-level agreement. The contribution of this study is a reproducible, component-space baseline that separates the learnability of the mechanistic response from site-specific noise and narrow target definitions, and that documents which components and operating regimes remain difficult even for the strongest raw surrogates (Viering & Loog, 2023).

The second study showed that predictive accuracy and physical admissibility are empirically distinct properties, and it provided a model-agnostic way to reconcile them. Every raw surrogate violated the stoichiometric conservation relation implied by the adopted model boundary, and several otherwise accurate surrogates also returned negative concentrations in low-value components. A positivity-preserving conservative

projection, applied after prediction and without retraining, returned every evaluated output to the declared invariant balance and the nonnegative component domain, removing both failure modes across all in-distribution and beyond-domain predictions with residuals at numerical precision (Kircher & Votsmeier, 2025). The study also established that projection is not a device for improving ordinary error, because its effect on aggregate accuracy was a measurable tradeoff that depended on the surrogate, the component, and the operating regime. The contribution is therefore twofold, combining a dependable enforcement layer with a paired evaluation logic in which admissibility diagnostics travel alongside accuracy metrics.

The third study integrated these requirements into a single surrogate design, Invariant-Constrained Second-Order Regression (ICSOR). Rather than repairing an opaque prediction after the fact, ICSOR learns a structured second-order regression head that keeps the engineering roles of operating variables, influent loads, nonlinear curvature, and cross-component coupling explicit, and it returns predictions through a checked deployment rule that guarantees exact conservation and nonnegativity. In the benchmark, ICSOR was the only pipeline whose returned component states satisfied both physical conditions on every test case, while its reported-composite accuracy remained within the leading group and only moderately above the best unconstrained learners. It was additionally the most sample-efficient model at the smallest dataset size and had the most favorable error profile across the full sample-size range, at the cost of higher fitting time and a dense but block-structured coefficient map. The contribution is a demonstration that an activated sludge surrogate can be constructed so that its output is simultaneously accurate, exactly admissible, and organized around an inspectable structure, rather than achieving these properties only in isolation (Rudin, 2019).

Taken together, the three studies reframe what it means for an activated sludge surrogate to be credible. Speed and low error are necessary but insufficient, because a usable surrogate should also return a component state that respects the conservation and nonnegativity structure of the underlying model and, where possible, expose the relationships it has learned (Rudin, 2019; Shen et al., 2023). The progression from raw benchmarking to post prediction enforcement to an integrated interpretable design provides both a set of concrete results and a transferable methodology. The same component-space evaluation, the same paired reporting of accuracy and admissibility, and the same constraint-aware model construction can be applied to other process systems whose surrogates must preserve conserved quantities, which positions the work as a contribution to physically grounded surrogate modeling as well as to activated sludge simulation (Karniadakis et al., 2021; Valente et al., 2025).

Outlook

The clearest direction for further work is validation beyond simulated data. The present evidence was obtained under a known mechanistic reference, a fixed component basis, and a defined constraint structure, which made it possible to isolate prediction error, conservation, and nonnegativity without confounding them with measurement noise. Extending the framework to full-scale wastewater treatment records will require attention to sensor error, calibration uncertainty, and unmodeled biological pathways, all of which complicate both the reference state and the invariant structure used for enforcement (Machado et al., 2014; Newhart et al., 2019). A natural first step is to test whether the admissibility guarantees remain meaningful when the conservation relation is estimated from imperfect plant data rather than from an exact stoichiometric description.

A second direction is the move from steady-state prediction toward transient and dynamic operation. Activated sludge plants experience diurnal and seasonal load variation, and many operational decisions depend on time-dependent behavior rather than on a single steady state. Adapting the surrogate benchmark, the projection step, and the checked deployment rule to dynamic trajectories would broaden their relevance to real-time monitoring, setpoint design, and advanced process control (Croll et al., 2023; Sadeghassadi et al., 2018). This extension raises open questions about how conservation and nonnegativity should be enforced along a trajectory rather than at a single operating point.

A third direction concerns the scope of the physical guarantees themselves. The conservation used here reflects the stoichiometric invariants of the adopted model within a declared reactor boundary, and it does not claim elemental closure, thermodynamic feasibility, kinetic realism, or plant-wide balance. Later work could enrich the feasible set with additional physically motivated constraints, couple the data driven surrogate with differentiable mechanistic components, and quantify the uncertainty of both the prediction and the correction so that admissible outputs also carry calibrated confidence (Shen et al., 2023; Valente et al., 2025). Strengthening the guarantee in these ways would make admissible surrogates more defensible for detailed design and regulatory reporting.

The methods can also be advanced on efficiency and scalability. The enforcement step and the ICSOR deployment optimizer added modest but nonzero cost, and the integrated model required more fitting time than the fastest baselines. Reducing simulation demand through active or sequential sampling, accelerating the constrained deployment for higher-dimensional component sets, and exploring richer coupling and curvature representations would improve the practicality of the approach for large screening and optimization workflows, where a surrogate is queried many times (Aboagye et al., 2021; Bhosekar & Ierapetritou, 2018; Durkin et al., 2024; Padron-Paez et al., 2020).

Improvements of this kind would help the framework scale to plant-wide models with more components and processes.

Finally, the results invite deeper integration with engineering decision-making. A surrogate that is fast, physically admissible, and structurally inspectable is well suited to serve inside optimization and control loops where each candidate solution must be both trustworthy and interpretable (Mao et al., 2024; Rudin, 2019; Wang et al., 2025). Later studies could examine how admissible and interpretable surrogates change the quality and acceptability of design and operating decisions relative to unconstrained black-box predictors, and how practitioners weigh a small, well-understood accuracy cost against exact physical dependability and structural transparency. Answering these questions would connect the methodological contributions of the research to the practical goal that motivates them, which is more reliable and more explainable support for the operation of biological wastewater treatment systems.

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Appendix A

Activated Sludge Component Basis and Composite Definitions

The three studies share a common representation of the activated sludge state defined by the Activated Sludge Model No. 2d with two-step nitrification (ASM2d-TSN), following the Activated Sludge Model family (Henze et al., 2015) and the two-step nitrification formulation of Guerrero et al. (2011). The effluent state is a vector of twenty components, comprising ten dissolved species and ten particulate species. Table A.1 lists the symbol, description, and native reporting unit of each component. The same basis defines both the supervised prediction targets and the space in which mass conservation and nonnegativity are assessed.

The reported water quality composites are not fitted directly. Chemical oxygen demand (COD), total nitrogen (TN), total phosphorus (TP), and total suspended solids (TSS) are computed from a predicted component vector through a fixed linear composition map, so that each composite is an interpretive summary of the underlying component state rather than an independent response. Chemical oxygen demand aggregates the oxygen-demand contributions of the soluble and particulate organic components, total nitrogen aggregates the inorganic and organically bound nitrogen contributions, total phosphorus aggregates the soluble, stored, precipitated, and biomass-bound phosphorus contributions, and total suspended solids aggregates the particulate components. Because each composite is a linear function of the component vector, any prediction that is admissible in the component space also yields internally consistent composite values (Henze et al., 2015).

Table A.1: Activated sludge component basis of the ASM2d-TSN state used across the studies.

Symbol	Description	Unit
<i>Dissolved components</i>		
S_O	Dissolved oxygen	g O ₂ /m ³
S_F	Fermentable, readily biodegradable substrate	g COD/m ³
S_A	Fermentation products (acetate)	g COD/m ³
S_{NH4}	Ammonium nitrogen	g N/m ³
S_{NO2}	Nitrite nitrogen	g N/m ³
S_{NO3}	Nitrate nitrogen	g N/m ³
S_{N2}	Dinitrogen	g N/m ³
S_{PO4}	Inorganic soluble phosphorus (phosphate)	g P/m ³
S_I	Inert soluble organic matter	g COD/m ³
S_{ALK}	Alkalinity (bicarbonate)	mol/m ³
<i>Particulate components</i>		
X_I	Inert particulate organic matter	g COD/m ³
X_S	Slowly biodegradable substrate	g COD/m ³
X_H	Heterotrophic biomass	g COD/m ³
X_{PAO}	Phosphate-accumulating-organism biomass	g COD/m ³
X_{PP}	Stored polyphosphate	g P/m ³
X_{PHA}	Stored polyhydroxyalkanoates	g COD/m ³
X_{AOB}	Ammonia-oxidizing biomass	g COD/m ³
X_{NOB}	Nitrite-oxidizing biomass	g COD/m ³
X_{MeP}	Precipitated metal phosphate	g TSS/m ³
X_{MeOH}	Metal hydroxide	g TSS/m ³

Appendix B

Surrogate Models and Shared Experimental Protocol

The comparative studies drew on a common set of statistical surrogates spanning the main families used for structured tabular regression. Table B.1 lists each surrogate, its family, and how it produced the twenty-component effluent prediction. The set combines boosted trees, randomized tree ensembles, kernel and neighbor methods, latent-variable and sparse multi-task regression, and neural networks, so that flexible nonlinear learners could be compared with simpler or more inspectable alternatives under a fixed data and scoring protocol (Arik & Pfister, 2021; Borisov et al., 2024; Breiman, 2001; Chen & Guestrin, 2016; Hornik et al., 1989; Wold et al., 2001). The interpretable Invariant-Constrained Second-Order Regression (ICSOR) surrogate developed in the third study was evaluated against comparators drawn from this same set.

A single mechanistic benchmark supported all comparisons. Steady-state cases were generated from the ASM2d-TSN simulator over a defined operating and influent domain using a seed-42 randomized Latin hypercube design (McKay et al., 1979), and each case was accepted only when its steady-state balance residual met a fixed numerical tolerance. Feature and target transformations were fitted on training partitions alone and inverted before scoring, which prevented leakage from held-out data (Pineiro et al., 2025). Hyperparameters were selected by seeded Optuna studies with a tree-structured Parzen estimator (Akiba et al., 2019), and the selected settings were held fixed across the nested and repeated sample-size analyses so that learning behavior reflected data quantity rather than repeated retuning (Viering & Loog, 2023). In-distribution performance was estimated by cross-validation with shared folds, and beyond-domain performance used separately simulated excursions outside the sampled envelope. This shared protocol makes the differences reported across the studies attributable to model family and physical enforcement rather than to inconsistent data handling.

Table B.1: Comparator surrogates evaluated across the studies and their effluent output handling.

Surrogate	Family	Effluent output handling
XGBoost	Boosting	One regressor per component
LightGBM	Boosting	One regressor per component
CatBoost	Boosting	One regressor per component
AdaBoost	Boosting	One regressor per component
Random Forest	Randomized tree ensemble	Native multi-output regression
Extra Trees	Randomized tree ensemble	Native multi-output regression
Support vector regression	Kernel method	One regressor per component
k -nearest neighbors	Neighbor method	Joint multi-output prediction
Partial least squares	Latent-variable regression	Joint twenty-component regression
Multi-task Elastic Net	Sparse multi-task regression	Joint twenty-component regression
Multi-task Lasso	Sparse multi-task regression	Joint twenty-component regression
Multilayer perceptron	Neural network	Joint twenty-component regression
TabNet	Neural network	Joint twenty-component regression

Appendix C

Evaluation Metrics and Physical Diagnostics

Accuracy was reported in two spaces, and physical behavior was quantified through two diagnostics. Accuracy in the component space used standardized errors so that components with different physical units could be combined, whereas accuracy in the reported-composite space used errors on chemical oxygen demand, total nitrogen, total phosphorus, and total suspended solids. Physical behavior was assessed before composite calculation so that numerical accuracy would not conceal an inadmissible component vector. Table C.1 summarizes each quantity, the space in which it is defined, and what it measures.

Component-space errors were normalized by the component standard deviations of the applicable training partition, which prevents high-magnitude components from dominating an aggregate score and keeps the in-distribution and beyond-domain results on a comparable scale. The macro-average coefficient of determination weights every component equally rather than by variance, so that easily predicted components do not conceal weak performance on more difficult ones. The mass-conservation residual measures the deviation of a prediction from the sample-specific stoichiometric invariants of the adopted model boundary, and the nonnegativity magnitude measures the total standardized negative mass in a prediction, so a component vector is physically admissible in the sense used here only when both quantities lie within numerical tolerance. The normalized area under the learning curve condenses an entire error-versus-sample-size trajectory into a single value, which allows sample efficiency to be compared without ranking models at only one training size (Viering & Loog, 2023).

Table C.1: Evaluation quantities used across the studies.

Quantity	Space	What it measures
Normalized mean squared, root mean squared, and mean absolute error	Component (standardized)	Average standardized component prediction error
Macro-average coefficient of determination	Component	Equal-weight variance explained across components
Root mean squared, mean absolute, mean absolute percentage, and mean squared error	Reported composite	Error on the COD, TN, TP, and TSS summaries
Coefficient of determination	Reported composite	Variance explained in the composite space
Mass-conservation residual	Component	Deviation from the stoichiometric invariants
Nonnegativity magnitude	Component	Total standardized negative component mass
Normalized area under the learning curve	Sample-size sweep	Sample-averaged error across training sizes

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List of Publications and Presentations

Publication

Selerio, E. F. (2026). An interpretable statistical surrogate of activated sludge systems that preserves mass conservation and component non-negativity. *Digital Chemical Engineering*, 20, Article 100329. doi: 10.1016/j.dche.2026.100329

Presentation

Selerio, E. F. (2023, April 21–22). *Mass transport, biofilm accumulation, and biotransformation in recycling down-flow hanging sponge systems* [Poster presentation]. 4th International Conference on Sustainable Energy Ecosystems (SEECON 2023), University of San Carlos–Talamban Campus, Cebu City, Philippines.